Does the Financial Accelerator accelerate inequalities?*

Francesco Ferlaino †

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Abstract

This study examines the redistribution effects of a conventional monetary policy shock among households in the presence of financial frictions on the production side of the economy. In this regard, I adopt a Heterogeneous Agents New Keynesian model that features a financial accelerator. The results show that with contractionary monetary policy, the Gini coefficient of wealth and other inequality measures are significantly higher in relative terms than in the counterfactual scenario with no frictions, proving that wealth inequality is substantially "accelerated" by firms' financial characteristics. Furthermore, consumption dynamics are also affected, suggesting that financial frictions have considerable effects on how households consume and save along their distribution, since they rely differently on labor income for consumption smoothing and have different marginal propensities to consume.

Keywords: heterogeneous agents, financial frictions, monetary policy, New Keynesian models, inequalities.

JEL codes: E12, E21, E44, E52, G51

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†DEMS, Università degli studi di Milano - Bicocca. Email: f.ferlaino@campus.unimib.it.

1 Introduction

Until quite recently, the redistribution effects of monetary policy have raised little concern among macroeconomists, especially those interested in theoretical models. However, the last decade has witnessed the growing attractiveness of this topic for several reasons. The Great Recession exacerbated the rise in wealth inequality that has been ongoing since the 1980s (for a review of inequality history, especially in advanced economies, see Piketty, 2017), coming to a point where ignoring it could translate into missing important aspects of the monetary transmission mechanism. For instance, Ahn et al. (2017) point out how the composition of micro data incorporated in a Dynamic Stochastic General Equilibrium (DSGE) model could have significant effects on macro aggregate fluctuations, and vice versa, how aggregate shocks affect micro features as well. Furthermore, the increasing power of computer processors and the development of new numerical techniques to manage large numbers of grid points have made possible to solve models featuring heterogeneity in a relatively short time; in the past, such models were highly time-consuming or simply impossible to solve. Fueled by these premises and building on frameworks developed by Huggett (1993), Aiyagari (1994) and Krusell and Smith (1998), a new strand of Heterogeneous Agents New Keynesian (HANK) models took off in recent years, trying to assess the effects that aggregate shocks of different kinds could have on the wealth distribution of households, and, on the other hand, how the shape of this distribution could affect the propagation of such shocks.

Contributing to this strand of the macroeconomic literature, this study assesses the importance of financial frictions on wealth distribution and household consumption for an economy subject to a contractionary conventional monetary policy shock. One of the main findings of Kaplan et al. (2018) is that, unlike in Representative Agent New Keynesian (RANK) models, most of the monetary policy transmission on households' consumption does not pass through direct effects (e.g., intertemporal substitution), but rather through indirect effects, such as labor dynamics, fiscal policy, and changes in asset prices.[1] More specifically, in their baseline model, labor income fluctuations are the most important component, accounting for more than half of the percentage change in aggregate consumption, leaving a marginal role for direct effects. In light of this result, I expect the presence of financial frictions in the production sector to be highly significant for wealth and consumption distribution shifts after a change in monetary policy, since these frictions affect the part of the economy where households are employed.

Standard New Keynesian models aimed at studying monetary policy usually ignore the financial structure of the production sector, in light of Modigliani and Miller (1958)

[1] Table 1 in Kaplan et al. (2018) displays how in standard RANK models, direct effects account for almost 100% of the monetary transmission. This percentage could drop up to 50% in a Two Agents New Keynesian (TANK) model, indicating that heterogeneity among households actually matters. Nonetheless, in TANK models, direct effects are still the most important ones.

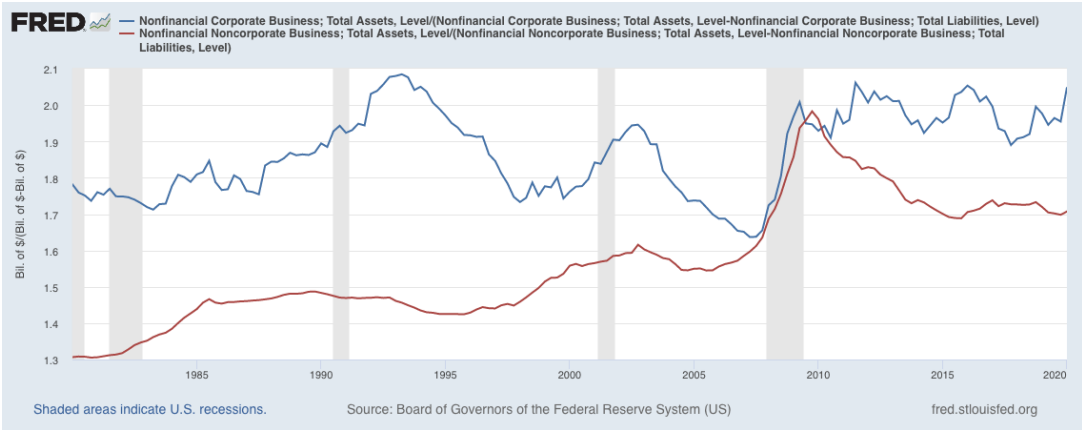


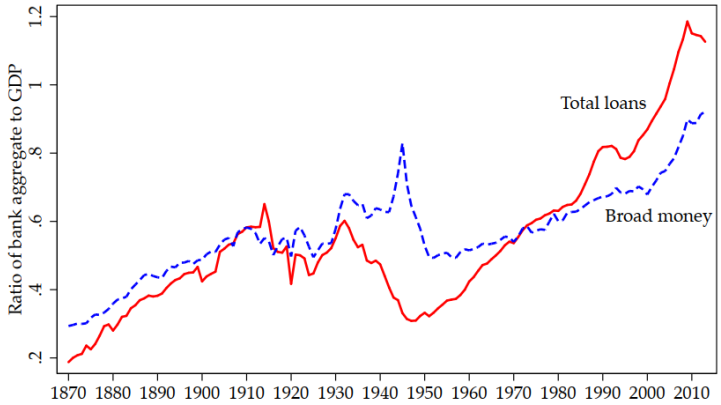
Figure 1: leverage level for non-financial firms in US

The blue line shows the leverage trend for non-financial corporate firms, and the red line shows that for non-financial non-corporate firms. Leverage is calculated as assets over equity, where equity is the difference between assets and liabilities at the aggregate level. Data are quarterly.

Source: Flow of Funds, National Accounts, Board of Governors of the Federal Reserve System (US).

theorem of capital structure irrelevance. However, over the years, an increasing number of economists have pointed out how leverage is significant for firms' decisions, and empirical data show a growing trend for aggregate debt in firms' balance sheet. Figure 1 displays the leverage evolution for non-financial corporate (blue line) and non-corporate (red line) firms in the US in the last 40 years. Corporate firms show leverage levels that are close to the value of two, rather than one (the case limit of no external debt). Although aggregate leverage declined just before the great financial crisis, it steeply increased when the crisis hit and remained at a high level. On the other hand, non-corporate firms used to rely less on external debt, but with a steadily increasing trend in leverage over time that burst with the 2008 financial crisis and started to revert immediately thereafter. However, since corporate firms' assets and liabilities represent the lion share in absolute terms, and since non-corporate leverage increased substantially over time, it is safe to assume that the aggregate leverage level was more than relevant over the past decades.

Jordà et al. (2017) use a macro-financial historical database covering 17 advanced economies over the last 150 years to show that the leverage level of the economy has become an important factor in explaining business cycles, making the role of financial variables crucial in understanding aggregate economic dynamics. Figure 2 displays the *financial hockey stick*, one of the main findings of their paper. Private loans (calculated as the ratio of private credit to GDP) took the spotlight over time in place of money. Although Jordà et al. (2017) emphasize the fact that the recent increase in leverage level was driven in large part by households borrowing and use a different measure of leverage from the one adopted in the model below, a final major suggestion by the authors is the increasing need for models that include financial dynamics, both on the household and firm side. In this respect, my work aims to embrace this proposition to study redistribution effects.



Notes: *Total loans* is bank lending to the non-financial private sector, *Broad Money* is M2 or similar broad measure of money, both expressed as a ratio to GDP averaged over the 17 countries in the sample. See text.

Figure 2: The financial hockey stick
Source: Jordà et al. (2017)

The HANK model adopted for this purpose features asset market incompleteness, idiosyncratic income risk, sticky prices, and a *financial accelerator* on the production side, as in Bernanke et al. (1999). This acceleration revolves around entrepreneurs who require external financing to reach the desired level of capital for good production. The cost of these funds is determined by the aggregate level of entrepreneurs' leverage: as leverage increases, external funds costs rise as well, since entrepreneurs need a relatively higher share of external funds and hence are more exposed to outer risks. An increase in these costs results in less capital, a decrease in goods production, and a lower level of output, which ultimately results in lower equity in the next period, generating the so-called financial accelerator of the economy. Including this mechanism in a model with household heterogeneity helps to assess the impact of this "acceleration" on wealth and consumption.

My main finding is that the financial accelerator is indeed an accelerator of inequality. After an increase in the nominal interest rate in the presence of borrowing frictions, the model estimates a Gini coefficient for wealth on impact almost 50% higher with respect to the counterfactual case in which the financial accelerator is shut down. A similar value, at least of impact, can be observed for the Gini coefficient of consumption. The intuition behind these results is that households respond differently along their wealth distribution. Households experiencing the highest shifts are those closer to the borrowing constraint, which is in line with recent findings in the HANK literature aiming to break the Permanent Income Hypothesis (PIH). Some agents are mostly (if not fully) dependent on their current income for consumption, and cannot smooth it in time because of their lack of savings.[2] The additional production depression brought by the financial accelerator has significant effects on outcome, labor and wages, which relatively

[2]As I explain later in Section 2, households cannot borrow resources to smooth consumption

hits harder households relying more on labor income than income coming from profits or interests on savings. Because more households are "pushed" to the constraint, the model generates greater wealth inequality. Eventually, this mechanism causes a greater decrease in consumption among these Hand-to-Mouth (HtM) households compared to the case with no financial accelerator. Interestingly, in the baseline model, aggregate consumption does not change much between the two scenarios, suggesting that wealthier households (who hold a greater amount of liquidity) gain more from an increase in the interest rate, enhancing consumption inequality.

As stated previously, my research lies in the rapidly growing literature on household heterogeneity within a New Keynesian framework. TANK models constitute a parsimonious yet powerful way to introduce household heterogeneity, with interesting results in monetary and fiscal policy evaluations (e.g., Galí et al., 2007; Bilbiie, 2008). Moreover, Debortoli and Galí (2017) showed how TANK models can reasonably approximate the predictions of a HANK model regarding the effects of an aggregate shock on aggregate variables. Nevertheless, they also point out that TANK models are not suitable for addressing other questions, such as the change in households' wealth distribution. Therefore, in such cases, we must resort to fully-fledged HANK models (e.g., Kaplan et al., 2018; Bayer et al., 2019; Auclert et al., 2021; Luetticke, 2021). Aside from the incredible contribution provided in creating algorithms that allow working with enormous amounts of grid points and, hence, with a great variety of households, a common goal is usually to match empirical micro data as accurately as possible. Differently, the aim of this study is to use a simpler model, taking advantage of such methodologies, to study the dynamics generated from the peculiar structure of the production sector in the economy.

My contribution is, obviously, also related to the financial frictions literature, in particular to the branch studying the financial accelerator generated by the existence of an "External Finance Premium" for firms (e.g., Carlstrom and Fuerst, 1997; Bernanke et al., 1999; Christiano et al., 2014; Carlstrom et al., 2016). In their seminal paper, Bernanke et al. (1999) articulate why the Modigliani and Miller (1958) assumption of financial structure irrelevance for real economic outcomes could be too limiting in certain cases, especially when frictions in financial markets are not small. The two main justifications the authors provide are that (i) even relatively small changes in entrepreneurial wealth could deliver important cyclical fluctuations (a line of thinking that goes back to Fisher, 1933) and (ii) empirical studies have appointed growing importance to credit market frictions, thereby increasing the need to fill the gap present in the theoretical literature at the time. My idea is to verify whether this acceleration mechanism takes place and has significant results also on inequality measures, a path that could not be explored by Bernanke et al. (1999) because they assumed a representative household in their model. Adding heterogeneity among households allows to understand not only how shocks affect the aggregate variables of the business cycle, but also the implications at idiosyncratic

levels.

To the best of my knowledge, to date, only a few studies consider financial frictions in a HANK environment, such as Guerrieri and Lorenzoni (2017), Nakajima and Ríos-Rull (2019) and Fernández-Villaverde et al. (2023). Nonetheless, none of these studies seem to focus on the inequality outcomes of conventional monetary policy. The paper by Lee et al. (2020) is probably the closest to my research. However, the substantial difference with my study is in the type of friction in place and, hence, in the dynamics that are meant to be comprehended. They build on the works of Gertler and Kiyotaki (2010) and Gertler and Karadi (2011) to analyze the effects of frictions on banks' balance sheets, the presence of which directly affects households' chances to borrow resources from financial intermediaries (higher borrowing rates) and, consequently, to smooth consumption. On the other hand, my study focuses more on *direct* frictions on production firms that have *indirect* effects on households, mainly through changes in labor income. Since a consistent share of households relies entirely (or for a good part) only on labor income for their consumption, it becomes important to study how inequalities are shaped not only when the banking sector is not running smoothly, but also when firms' financing becomes more costly because of their financial structure.

The remainder of this paper is organized as follows. Section 2 outlines the model. Section 3 depicts the solution method for the model. Section 4 explains the calibration strategy. Section 5 displays quantitative and empirical results. Section 6 gives summary conclusions.

2 The model

The model comprises households, financial intermediaries, a production sector, a central bank, and the government. Households consume, earn income (either from labor or profit, depending on the household type), and save in a liquid asset, which yields an interest rate. Financial intermediaries obtain deposits from households and lend them to the production sector, which, in turn, is responsible for the production of goods and capital. The central bank is in charge of monetary policy and sets the nominal interest rate, whereas the government acts as fiscal authority and chooses how to finance government spending. Time is discrete and infinite. The behavior of each agent is explained in detail below.

2.1 Households

There is a continuum of ex ante identical households of measure one indexed by $i \in [0, 1]$. They are infinitely lived, have time-separable preferences with time-discount factor β and their utility function u is affected positively by consumption, c_{it}, and negatively by labor,

l_{it}, with $l_{it} \in [0,1]$ being hours worked as a fraction of the time endowment, normalized to 1. The utility function u is strictly increasing and strictly concave in consumption and strictly decreasing and strictly convex in labor. Household i value function is the following:

$$V = E_0 \max_{\{c_{it}, l_{it}\}} \sum_{t=0}^{\infty} \beta^t u(c_{it}, l_{it}) \ , \tag{1}$$

where I assume households have separable preferences with a Constant Relative Risk Aversion (CRRA) form:

$$u(c,l) = \frac{c^{1-\xi}}{1-\xi} - \psi \frac{l^{1+1/\nu}}{1+1/\nu} \ . \tag{2}$$

There are two types of household: workers and rentiers. Workers supply labor, l_{it}, in the production sector and have positive idiosyncratic labor productivity, $h_{it} > 0$. Because the global wage level, W_t, is the same for everyone, their income is given by $W_t h_{it} l_{it}$. Rentiers have zero labor productivity, $h_{it} = 0$, but collect a proportional share of total profits generated from the production sector, Π_t. Idiosyncratic labor productivity h_{it} follows an exogenous Markov chain according to the following first-order autoregressive process and a fixed probability of transition between the worker and the rentier state:

$$h_{it} = \begin{cases} exp(\rho_h log(h_{it-1}) + \epsilon_{it}^h) & \text{with probability } 1-\zeta \text{ if } h_{it-1} \neq 0 \\ h_t^H & \text{with probability } \iota \text{ if } h_{it-1} = 0 \\ 0 & \text{else} \end{cases} \tag{3}$$

where $\epsilon_{it}^h \sim N(0, \sigma_h)$ and h_t^H is the highest possible productivity realization for workers. The parameter $\zeta \in (0,1)$ is the probability that a worker becomes a rentier, while $\iota \in (0,1)$ is the probability that a rentier becomes a worker. As stated above, workers who become rentiers leave the labor market ($h_{it} = 0$), whereas rentiers that become workers are endowed with productivity h_t^H.[3] Workers and rentiers pay the same level of taxation, τ, on their income.

The asset market is incomplete: there are no Arrow-Debreu state-contingent securities, households self-insure themselves only through savings in a non-state contingent risk-free liquid asset, a_{it}, and they cannot get indebted on that, that is, an *ad hoc* borrowing constraint exists ($a_{it} \geq 0$). Thus, households cannot borrow from financial intermediaries to smooth their consumption. The household's budget constraint is:

$$c_{it} + a_{it+1} = \left(\frac{R_t}{\pi_t}\right) a_{it} + (1-\tau)(W_t h_{it} l_{it} + \mathbf{I}_{h_{it}=0} \Pi_t) \ , \tag{4}$$

where $\mathbf{I}_{h_{it}=0}$ takes the value of 1 if the household is a rentier and 0 otherwise. On the

[3] Appendix A contains details on the transition matrix for household productivity.

right-hand side, we have households' expenditure, that is, consumption, c_{it} and 1-year-maturity savings, a_{it+1}. The left-hand side corresponds to households' total earnings: work/rent income net of taxes, $(1-\tau)(W_t h_{it} l_{it} + \mathbf{I}_{h_{it}=0}\Pi_t)$, and the gross real interest rate on previous savings, $(R_t/\pi_t)a_{it}$.

Liquid assets held by households are a mix of deposits, D_t, and government bonds, B_t, so that we have the following relation:

$$A_t = D_t + B_t \ , \tag{5}$$

where $A_t = \int a_{it} di$. Deposits and bonds are perfect substitutes, which means that they carry the same real interest rate, $\frac{R_t}{\pi_t}$, and that households are indifferent to the composition of A_t.[4]

Two Bellman equations characterize the household dynamic planning problem. The first one is for the case where households are workers at time t and thus they take expectations according to the following:

$$\mathbf{V}_i^w(a_{it}, h_{it}, S_t) = \max\left\{u(c_{it}, l_{it}) + \beta E_t[(1-\zeta)\mathbf{V}_t^w(a_{it+1}, h_{it+1}, S_{t+1}) + \zeta \mathbf{V}_t^r(a_{it+1}, S_{t+1})]\right\} \ . \tag{6}$$

The second reflects the rentier household point of view at time t:

$$\mathbf{V}_i^r(a_{it}, S_t) = \max\left\{u(c_{it}, l_{it}) + \beta E_t[(1-\iota)\mathbf{V}_t^r(a_{it+1}, S_{t+1}) + \iota \mathbf{V}_t^w(a_{it+1}, h_{it+1}, S_{t+1})]\right\} \ , \tag{7}$$

with S_t being a vector of aggregate state variables.

2.2 Financial intermediaries

Financial intermediaries collect deposits from households and promise returns equal to the real risk-free interest rate, R/π. For ease of display, I assume that the production sector is run by entrepreneurs, who are a mass-zero group of managers who are entitled to all the profits generated in the production sector and rebate them to rentier households. Financial intermediaries and entrepreneurs are responsible for the financial frictions considered in this model. Following Bernanke et al. (1999), I assume a continuum of entrepreneurs, indexed by j. Entrepreneur j acquires capital, K_j, from capital producers at the end of period t that is used at time $t+1$. To buy capital for production, entrepreneurs rely on two type of financing: internal financing, that is, equity, N_j, and external financing, D_j.

Entrepreneur j balance sheet at period $t+1$ is:

[4] I assume that each household has the same portfolio composition of liquid assets, which is equal to their aggregate level.

$$q_t K_{jt+1} = N_{jt+1} + D_{jt+1} \ , \tag{8}$$

where q is the price of capital at the time of the purchase.

One prerequisite for the functioning of this financial accelerator is that entrepreneurs are not indifferent to the composition of their balance sheet; that is, external financing is more expensive than internal financing. To introduce this feature, a Costly State Verification (CSV) problem à la Townsend (1979) exists, in which lenders (i.e., financial intermediaries) must pay a fixed auditing cost to observe the realized returns of borrowers (i.e., entrepreneurs). A relatively higher demand for debt increases auditing costs, resulting in a lower level of aggregate capital obtained for production.

Entrepreneurs repay investment banks with a portion of their realized returns on capital. In this framework, entrepreneurs are risk-neutral, while households are risk-averse. This implies a loan contract in which entrepreneurs absorb any aggregate risk on the realization of their profits. I assume the existence of an idiosyncratic shock to entrepreneur j, ω_j,[5] on the gross return on aggregate capital, R^K. The idiosyncratic shock ω has a log normal distribution of mean $E(\omega) = 1$ that is i.i.d. across time and across entrepreneurs, with a continuous and once differentiable c.d.f., $F(\omega)$.[6]

The optimal contract for financial intermediaries is:

$$\bar{\omega}_{jt+1} R^K_{t+1} q_t K_{jt+1} = Z_{jt+1} D_{jt+1} \ , \tag{9}$$

where Z_j is the gross non-default loan rate and $\bar{\omega}_j$ is the threshold value for entrepreneur j such that, for $\omega_{jt+1} \geq \bar{\omega}_{jt+1}$, entrepreneur j repays $Z_{jt+1} D_{jt+1}$ to financial intermediaries and retains $\omega_{jt+1} R^K_{t+1} q_t K_{jt+1} - Z_{jt+1} D_{jt+1}$. In the case of $\omega_{jt+1} < \bar{\omega}_{jt+1}$, instead, she cannot repay and defaults on her debt, obtaining nothing. Since entrepreneurs' future realizations of capital returns are only known by entrepreneurs ex-post, financial intermediaries must pay a fixed auditing cost, μ, to recover what is left of entrepreneur j's activity after default, obtaining $(1-\mu)\omega_{jt+1} R^K_{t+1} q_t K_{jt+1}$.

Because of the optimal contract, financial intermediaries should receive an expected return equal to the opportunity cost of their funds. By assumption, they hold a perfectly safe portfolio (i.e., they are able to perfectly diversify the idiosyncratic risk involved in lending), and the opportunity cost for financial intermediaries is the real gross risk-free rate, R/π. It follows that the participation constraint for financial intermediaries that must be satisfied in each period $t+1$ is:

[5] As noted by Christiano et al. (2014), ω could be thought of as the idiosyncratic risk in actual business ventures: in the hands of some entrepreneurs, a given amount of raw capital is a great success, while in other cases may be not.

[6] Appendix B.1 provides analytical expressions for $F(\omega)$ and other functions used in the following equations.

$$[1 - F(\bar{\omega}_{jt+1})]Z_{jt+1}D_{jt+1} + (1-\mu)\int_0^{\bar{\omega}_{jt+1}} \omega_j dF(\omega_j) R^K_{t+1} q_t K_{jt+1} \geq \frac{R_{t+1}}{\pi_{t+1}} D_{jt+1} \ , \quad (10)$$

where $F(\bar{\omega}_j^F)$ is entrepreneur j default probability. Since financial markets are in perfect competition, (10) must hold with equality. The first term on the left-hand side of (10) represents the revenues received by financial intermediaries from the fraction of entrepreneurs that do not default, whereas the second term is what financial intermediaries can collect from defaulting entrepreneurs after paying monitoring costs.

Following the notation proposed in Christiano et al. (2014), I combine (8), (9), and (10) to write the following relationship:

$$EFP_{jt+1} = f\left(\bar{\omega}_{jt+1}, LEV_{jt+1}\right) \ , \ \text{with} \ f'(LEV_{jt+1}) > 0 \ . \quad (11)$$

where EFP is the "External Finance Premium" that Bernanke et al. (1999) define as the ratio between the return on capital and the real risk-free rate, $R^K/(R/\pi)$, and $LEV = qK/N$ is entrepreneur's leverage. The EFP can be considered a measure of the cost of external funds for the entrepreneur and, therefore, as a proxy for the strength of financial frictions. The $(\bar{\omega}_{jt+1}, LEV_{jt+1})$ combinations that satisfy (11) define a menu of state $(t+1)$-contingent standard debt contracts offered to entrepreneur j, who chooses the contract that maximizes its objective.

In Appendix B.2, I illustrate the entrepreneur j's optimization problem, which provides three important outcomes. First, the EFP increases monotonically with LEV. This means that entrepreneurs with a higher level of leverage pay a higher EFP. Second, the threshold value for entrepreneur j's default, $\bar{\omega}_j$, is endogenously defined by the EFP. Third, the fact that $\bar{\omega}_j$ depends only on the aggregate variables (R,R^K and π) implies that every entrepreneur will choose the same firm structure, that is, $\bar{\omega}$ and LEV. Therefore, it is possible to drop the superscript j in the notation and consider a representative entrepreneur.

The other fundamental equation for the functioning of this financial accelerator is the law of motion for entrepreneurs' equity, which is expressed as follows:

$$N_{t+1} = \gamma \left[q_{t-1} R^K_t K_t - \frac{R_t}{\pi_t} D_t - \mu G(\bar{\omega}_t) q_{t-1} R^K_t K_t \right] \ . \quad (12)$$

Equation (12) states that entrepreneurs' equity after the production process at time t is equal to the gross return on capital net of the loan repayment and auditing costs (which are borne by entrepreneurs because they are risk-neutral). Parameter γ represents the share of surviving entrepreneurs who bring their equity to the production process from one period to the next. Conversely, the share of entrepreneurs $1-\gamma$ dies and consumes equity at time t (we can think of this as entrepreneurial consumption). As explained by

Carlstrom et al. (2016), this assumption avoids excessive entrepreneurs' self-financing in the long run.

Note that in (12) I did not included entrepreneurial labor, as usual in the literature (e.g., Bernanke et al. (1999), Christiano et al. (2014)). The assumption of entrepreneurial labor was introduced mainly to justify the initial amount of equity for new entrepreneurs that take the place of the dead ones. However, to keep the model as simple as possible, I follow Carlstrom et al. (2016), assuming that new entrepreneurs' initial equity comes from a lump-sum transfer from existing entrepreneurs. Even so, since the funding can be arbitrarily small and since only aggregate equity matters, this transfer can be neglected in equation (12).[7] Alternatively, (12) can be written in a more compact form as:

$$N_{t+1} = \gamma \left[1 - \Gamma(\bar{\omega}_t)\right] R_t^K q_{t-1} K_t \ , \tag{13}$$

where $[1 - \Gamma(\bar{\omega}_t)]$ is the share of capital returns to which non-defaulting entrepreneurs are entitled.[8] Equation (13), together with (11), explains the financial accelerator mechanism. Equation (11) states that an increase in entrepreneurs' leverage increases also the EFP. At the same time, (13) tells that an increase in the EFP increases $\bar{\omega}$ as well, negatively affecting entrepreneurs' equity level for the next period and, therefore, impacting the aggregate leverage.

2.3 Intermediate-good producers

Intermediate-good producers adopt a standard Cobb-Douglas production function with constant returns to scale, employing aggregate capital, K, supplied by entrepreneurs and labor, L, from workers:

$$Y_t = z_t L_t^{\alpha} K_t^{1-\alpha} \ , \tag{14}$$

where z represents the Total Factor Productivity (TFP).

TFP follows a first-order autoregressive process of type:

$$log(z_t) = \rho_z log(z_{t-1}) + \epsilon_t^z \ , \tag{15}$$

with ϵ_t^z following a normal distribution with mean 0 and variance σ^z.

Intermediate-good producers sell their production to resellers at a relative price MC_t. Therefore, their profit optimization is given by:

$$\Pi_t^{IG} = MC_t z_t L_t^{\alpha} K_t^{1-\alpha} - W_t L_t - r_t^K K_t \ . \tag{16}$$

[7] Bernanke et al. (1999) keep the share of income going to entrepreneurial labor at a very low level (on the order of 0.01), therefore neglecting this income sounds as a reasonable model simplification.

[8] See Appendix B.2

Since they are in perfect competition, their profit optimization problem returns the wage paid per unit of labor and the rent paid per unit of capital:

$$W_t = \alpha MC_t z_t \left(\frac{K_t}{L_t}\right)^{(1-\alpha)} , \tag{17}$$

$$r_t^K = (1-\alpha)\, MC_t z_t \left(\frac{L_t}{K_t}\right)^{\alpha} . \tag{18}$$

2.4 Resellers

Resellers are agents assigned to differentiate intermediate goods and set prices. Following Bayer et al. (2019), I assume that price adjustment costs follow a Rotemberg (1982) setup and that resellers are directly run by entrepreneurs, preserving their characteristics.[9] The demand for the differentiated good g is:

$$y_{gt} = \left(\frac{p_{gt}}{P_t}\right)^{-\eta} Y_t , \tag{19}$$

where $\eta > 1$ is the elasticity of substitution and p_g is the price at which good g is purchased.

Given (19) and the quadratic costs of price adjustment, resellers maximize:

$$E_0 \sum_{t=0}^{\infty} \beta^t Y_t \left\{ \left(\frac{p_{gt}}{P_t} - MC_t\right) \left(\frac{p_{gt}}{P_t}\right)^{-\eta} - \frac{\eta}{2\kappa} \left(log \frac{p_{gt}}{p_{gt-1}}\right)^2 \right\} , \tag{20}$$

with a time-constant discount factor.[10]

The New Keynesian Phillips Curve (NKPC) derived from the F.O.C. for price setting is as follows:

$$log(\pi_t) = \beta E_t \left[log(\pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right] + \kappa \left(MC_t - \frac{\eta - 1}{\eta} \right) , \tag{21}$$

where π_t is the gross inflation rate defined as $\frac{P_t}{P_{t-1}}$.

[9] Bayer et al. (2019) make the further assumption that price setting is delegated to a mass-zero group of households (managers) that are risk neutral and compensated by a share in profits. Since in my model the whole production sector is run by entrepreneurs that, by assumption, are risk neutral and entitled to all the profits generated in this sector, I do no need to make this further assumption.

[10] As explained by Bayer et al. (2019), only the steady-state value of the discount factor matters in the resellers' problem, due to the fact that I calibrate to a zero inflation steady-state, the same value for the discount factor of managers and households and approximate the aggregate dynamics linearly. This assumption simplifies the notation, since fluctuations in stochastic discount factors are virtually irrelevant.

2.5 Capital producers

After production at time t, entrepreneurs sell depreciated capital to capital producers at a price q_t. They refurbish depreciated capital at no cost,[11] and uses goods as investment inputs, I_t, to produce new capital, $\Delta K_{t+1} = K_{t+1} - K_t$, subject to quadratic adjustment costs. Finally, they resell the newly produced capital to entrepreneurs before entering the next period (therefore still at price q_t).

The law of motion for capital producers is:

$$I_t = \Delta K_{t+1} + \frac{\phi}{2}\left(\frac{\Delta K_{t+1}}{K_t}\right)^2 K_t + \delta K_t \ . \tag{22}$$

where δ is the depreciation rate for capital.

Then, capital producers maximize their profit, $q_t \Delta K_{t+1} - I_t$, w.r.t. newly produced capital, ΔK_{t+1}. This optimization problem delivers the optimal capital price:

$$q_t = 1 + \phi \frac{\Delta K_{t+1}}{K_t} \ . \tag{23}$$

This ensures that if the level of aggregate capital increases over time, so does its price.

It follows that entrepreneurs' return on capital does not depend only on goods production, but also on fluctuations of the capital price. Since entrepreneurs buy capital at the end of the period, they see that their capital at the beginning of the next period appreciated (depreciated) if q increases (decreases) relative to the past. The gross return on capital employed at time t can be written as:

$$R_t^K q_{t-1} K_t = r_t^K K_t + q_t K_t (1 - \delta) \ , \tag{24}$$

where the first term on the right-hand side is the marginal productivity of capital derived in (18) and the second term represents the eventual capital gain (or loss) net of capital depreciation.

I can rearrange and finally derive the gross interest rate of capital as:

$$R_t^K = \frac{r_t^K + q_t(1-\delta)}{q_{t-1}} \ . \tag{25}$$

2.5.1 Final-good producers

Final-good producers are perfectly competitive, buy differentiated goods from resellers at a given price, and produce a single homogeneous final good used for consumption,

[11] The "no cost" assumption does not mean that δK is refurbished for free. Capital producers still need to buy the exact amount of I necessary to refurbish depreciated capital, but do not waste any further resources in this process. In fact, the law of motion for capital producers in the steady-state (when $\Delta K = 0$) is $I = \delta K$.

government spending, and investment. The optimization problem of final-good producers is:

$$\max_{\{Y_t, y_{gt} \in [0,1]\}} P_t Y_t - \int_0^1 p_{gt} y_{gt} dg \ , \tag{26}$$

subject to the following Constant Elasticity of Substitution (CES) function:

$$Y_t = \left(\int_0^1 (y_{gt})^{\left(\frac{\eta-1}{\eta}\right)} dg \right)^{\left(\frac{\eta}{\eta-1}\right)} \ . \tag{27}$$

From the zero-profit condition, the price index of the final good is:

$$P_t = \left(\int_0^1 (p_{gt})^{(1-\eta)} \right)^{\left(\frac{1}{1-\eta}\right)} \ . \tag{28}$$

2.6 Central bank

The central bank is responsible for the monetary policy. It sets the gross nominal risk-free interest rate, R, reacting to the deviation from steady-state inflation, and engages interest rate smoothing. The Taylor-type rule employed by the central bank is as follows:

$$\frac{R_{t+1}}{\bar{R}} = \left(\frac{R_t}{\bar{R}} \right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_R)\rho_\pi} \epsilon_t^R \ , \tag{29}$$

where ϵ_t^R is the monetary policy shock defined as $log(\epsilon_t^R) \sim N(0, \sigma_R)$. The parameter $\rho_R \geq 0$ rules interest rate smoothing (if $\rho_R = 0$, the next-period interest rate depends only on inflation), whereas ρ_π captures the magnitude of the central bank's response to inflation fluctuations: the larger ρ_π, the stronger the central bank reaction (for the case limit $\rho_\pi \to \infty$, inflation is perfectly stabilized at its steady-state level).

2.7 Government

The government acts as fiscal authority. It determines the level of government spending, G_t, tax revenues, T_t and issuance of new bonds, B_{t+1}. Its budget constraint is given by:

$$B_{t+1} = \left(\frac{R_t}{\pi_t} \right) B_t + G_t - T_t \ , \tag{30}$$

where T_t are taxes collected from both worker and rentier households:

$$T_t = \tau \left(W_t h_{it} l_{it} + \mathbf{I}_{h_{it}=0} \Pi_t \right) \ . \tag{31}$$

Government bond issuance is regulated by the following rule:

$$\frac{B_{t+1}}{\bar{B}} = \left(\frac{B_t \frac{R_t}{\pi_t}}{\bar{B} \frac{\bar{R}}{\bar{\pi}}} \right)^{\rho_B} . \tag{32}$$

The parameter ρ_B captures how fast the government wants to balance its budget. When $\rho_B \to 0$, the government balances its budget by adjusting its spending. However, when $\rho_B \to 1$, the government is willing to roll over most of its outstanding debt.

2.8 Market clearing

Let $\Theta_t(a, h)$ be the joint distribution of liquid assets and productivity across households on date t.

The labor market clears when:

$$\int h \, l^*(a, h) \Theta_t(a, h) da dh = L_t \ , \tag{33}$$

where $l^*(a, h)$ is the optimal labor supply policy function of the household.

The liquid asset market clears when:

$$\int a^*(a, h) \Theta_t(a, h) da dh = A_t \ , \tag{34}$$

where $a^*(a, h)$ is the optimal saving policy function of the household.

The market for capital clears for (22) and (23).

Finally, good market clearing, which holds by Walras' law when other markets clear, is defined as:

$$Y_t = C_t + G_t + I_t + C_t^E + \mu G(\bar{\omega}_t) R_t^K q_{t-1} K_t \ , \tag{35}$$

where on the left-hand side we have total output. On the right-hand side, apart from household consumption, public expenditure and investments, we also find entrepreneurial consumption, C^E (provided by dying entrepreneurs), and auditing costs.[12]

3 Numerical implementation

To solve the model, I follow the solution proposed in Bayer et al. (2019) and Luetticke (2021). As the joint distribution, Θ_t, is an infinite-dimensional object (and therefore not computable), it is discretized and represented by its histogram, a finite-dimensional object. I solve the household's policy function using the Endogenous Grid-point Method

[12] Similarly to Kaplan et al. (2018), we can think of this last term as "financial services".

(EGM) developed by Carroll (2006), iterating over the first-order condition and approximating the idiosyncratic productivity process using a discrete Markov chain with three states using the Tauchen (1986) method. The log grid for liquid assets comprises of 100 points. I solve for aggregate dynamics by first-order perturbation around the steady-state, as in Reiter (2009). Differently from Bayer et al. (2019) and Luetticke (2021), I do not need to resort to the dimensionality reduction developed by Bayer and Luetticke (2020), because my joint distribution is represented by a bi-dimensional matrix (capital K does not display heterogeneity) with a total of 300 grid points, maintaining a sufficiently low computational time.

4 Calibration

The model is calibrated on the US economy, and because the focus is on conventional monetary policy, business cycle moments are targeted on the *Great Moderation* (i.e., 1985-2007). Periods in the model represent quarters; consequently, the following values for the calibrated parameters are intended quarterly unless otherwise specified. Table 1 provides a list of calibrated parameters for the model, whereas Table 2 shows the model's effectiveness in replicating wealth distribution and business cycle moments.

4.1 Households

For the households' utility function, I assume the coefficient of relative risk aversion $\xi = 2$, which is consistent with the findings of Attanasio and Weber (1995) and already used by Auclert et al. (2021). I set the Frisch elasticity of labor supply $\nu = 1$, in line with the results of Chetty et al. (2011). The parameter for the disutility of labor, ψ, is set to 5.5, to have an average value for hours worked equal to 1/2, as in Kaplan et al. (2018). The intertemporal discount factor, β, is equal to 0.987, so savings in deposits by households are sufficient to have a leverage for entrepreneurs of 2, the same value used by Bernanke et al. (1999) in their model, and a fair calibration given the data illustrated in Figure 1. I decide on purpose to impose a non-borrowing condition for households, setting the borrowing limit for liquidity $\underline{a} = 0$, to highlight the transmission mechanism of monetary policy through financial frictions on the production sector rather than on the lending sector.[13]

The calibration of the productivity transition matrix, which determines how households move between the worker and rentier states, aims to provide a distribution of wealth consistent with empirical data. As in Luetticke (2021), I assume that the probability of

[13]The lack of a negative ad hoc borrowing constraint denies a further instrument of parameterization, since in the literature this feature is often used to target the share of HtM or borrowing households. Nonetheless, the share of zero-wealth households generated by the model is still significant, approximately 16%.

Table 1: Calibrated parameters

Parameter	Value	Description
β	0.987	Discount factor
ξ	2	Relative risk aversion
ν	1	Frisch elasticity of labor
ψ	5.5	Disutility of labor
$\underline{a}$	0	Borrowing constraint
ι	0.0625	Prob. of leaving entr. state
ζ	0.0005	Prob. become rentier
ρ_h	0.98	Persistence of idio. prod. shock
σ_h	0.06	SD if idio. prod. shock
α	0.7	Labor share of production
δ	0.2	Depreciation rate
η	20	Elasticity of substitution
κ	0.09	Price stickiness
ϕ	5	Adjustment cost of capital
μ	0.12	Auditing costs
σ_ω	0.27	SD of the id. shock on entr.
γ	0.985	Entr. surviving rate
ρ_z	0.95	TFP shock persistence
σ_z	0.00915	TFP shock SD
R	1.0063	Nominal int. rate
ρ^R	0.8	Int. rate smoothing
ρ^π	1.5	Reaction to inflation
σ_R	0.0009	Monetary shock SD
τ	0.3	tax rate
ρ_B	0.86	Auto-correlation of debt

Table 2: Wealth distribution and business cycle moments

Wealth distribution moments

Target	Model	Target
Gini wealth	0.78	0.78
top 10% wealth	0.71	0.67
zero-wealth HHs	0.16	0.20∼0.30

Business cycle moments

Target	Model	Target
SD of Y (%)	1.38	1.38
σ^I/σ^Y	4.5	4.5
SD of C (%)	0.47	0.98
Corr. of Y with Y	1	1
Corr. of I with Y	0.99	0.92
Corr. of C with Y	0.95	0.92

Real GDP, investment and consumption are in logs. All data for business cycle moment analysis are processed with a H-P filter with $\lambda = 1600$. The calibrated moments for wealth distribution is the Gini index for wealth. For business cycle moments, SD of Y and SD of I after a TFP shock.

becoming a rentier is the same for workers independent of their labor productivity, and that once they become workers again, they start with the highest productivity realization. The probability of leaving the rentier state is $\iota = 0.0625$, following the findings of Guvenen et al. (2014) on the probability of dropping out of the top 1% income group in the US. The probability of moving from the worker to the rentier state is $\zeta = 0.0005$, a value calibrated to obtain a Gini coefficient for wealth of 78%, in line with empirical data from the Survey of Consumer Finances (SCF) 1983–2007 (Luetticke, 2021), implying a share of rentier households of approximately 1%. Regarding idiosyncratic income risk for labor productivity, I set autocorrelation $\rho_h = 0.98$ and standard deviation $\sigma_h = 0.06$, as estimated by Bayer et al. (2019).

4.1.1 Financial Intermediaries

The parameters concerning financial frictions on firms are in the ballpark of Bernanke et al. (1999) calibrations; therefore, the auditing cost is $\mu = 0.12$ and the standard deviation of the idiosyncratic shock on the entrepreneur's returns is $\sigma_\omega = 0.27$, which are calibrated to have $EFP_t = 1.005$ (and, therefore, a credit spread of 2% p.a.) when the corporate leverage is 2. The share of surviving entrepreneurs, γ, is calibrated such that, at steady-state, the equity level in (13) is equal to the equity implied by (11).

4.1.2 Production Sector

The labor share of production (accounting for profits) and capital depreciation rate follow standard values in the literature and are set respectively to $\alpha = 0.7$ and $\delta = 2\%$. The mark-up is also standard, at 5%, which implies elasticity of substitution between goods varieties $\eta = 20$. The price stickiness parameter in the NKPC, $\kappa = 0.09$, is calibrated to generate a slope of the curve similar to the one that would arise in a model with sticky prices à la Calvo, with an average price duration of four quarters. The adjustment cost of the capital parameter is calibrated to $\phi = 5$ to match investment-to-output volatility $\sigma(I)/\sigma(Y) = 4.5$ after a TFP shock (in line with empirical data for the US elaborated by Bayer et al., 2019) in the model where the financial accelerator is in place. The persistence of the TFP shock is $\rho_z = 0.95$, while the standard deviation $\sigma_z = 0.00915$ is calibrated to match the standard deviation of the US output (after HP filtering) in the targeted time period.

4.2 Central Bank and Government

Inflation at the steady-state is set to 0% per annum, and the nominal (therefore real) interest rate on bonds is 2.5%, a value in line with the real average federal funds rate for the *Great Moderation* period. I impose the same interest rate on all types of liquid savings (i.e., bonds and deposits); otherwise, households would choose to invest only in one asset

or the other. Regarding the Taylor rule adopted by the Central Bank, the parameter for interest rate smoothing is $\rho_R = 0.8$, according to the findings of Clarida et al. (2000), whereas the reaction to inflation fluctuations from the steady-state is $\rho_\pi = 1.5$, which is a common value in the macroeconomic literature. For the monetary policy shock magnitude, I follow Luetticke (2021) and assume that the central bank raises the nominal interest rate by 36 basis points annually, a shock size which is in line with findings from the empirical literature (e.g., Wieland and Yang, 2016).

The taxes set by the government are proportional to both labor income and profits, with a tax rate $\tau = 0.3$ that targets the ratio of government spending to GDP to a standard value in the New Keynesian literature, $G/Y = 20\%$. Since I am using a fiscal policy rule similar to the one adopted by Bayer et al. (2019), I also follow their estimation and set $\rho_B = 0.86$. This implies that the fiscal dynamic passes through government debt, with public spending adjusting to re-stabilize debt to its steady-state level.

5 Results

First, I show some aggregate fluctuations. This helps assess the consistency of the results with respect to Bernanke et al. (1999) findings. Subsequently, I study inequalities in the model, which are at the core of this research.

5.1 Aggregate fluctuations

As explained above, I assume that the economy is subject to an unexpected increase in the nominal interest rate in the first period (one-time innovation). Figure 3 shows the response of several aggregate variables to this shock, comparing the case when financial frictions are in place (blue line) or not (red line).[14] The financial accelerator for aggregate output and investment is also confirmed in models with household heterogeneity. Similarly to the top panel of Figure 3 in Bernanke et al. (1999), output and investments responses under financial frictions exhibit higher magnitude on impact and higher persistence over time.[15] On the other hand, fluctuations in aggregate consumption do not differ significantly between the two scenarios on impact. This result is consistent with the findings in Bernanke et al. (1999) as well.[16]

[14]Since the analysis of aggregate fluctuation is not at the core of this research, I decided to show in the main text impulse responses for three variables only. More aggregate IRFs can be found in Appendix C.

[15]Obviously, since in Bernanke et al. (1999) we have a fall in the nominal interest rate, the two dynamics are mirrored.

[16]The authors do not show impulse responses for consumption in their paper. Nonetheless, using replication codes as the one present in the Macroeconomic Model Data Base (https://www.macromodelbase.com) allows to see also this result.

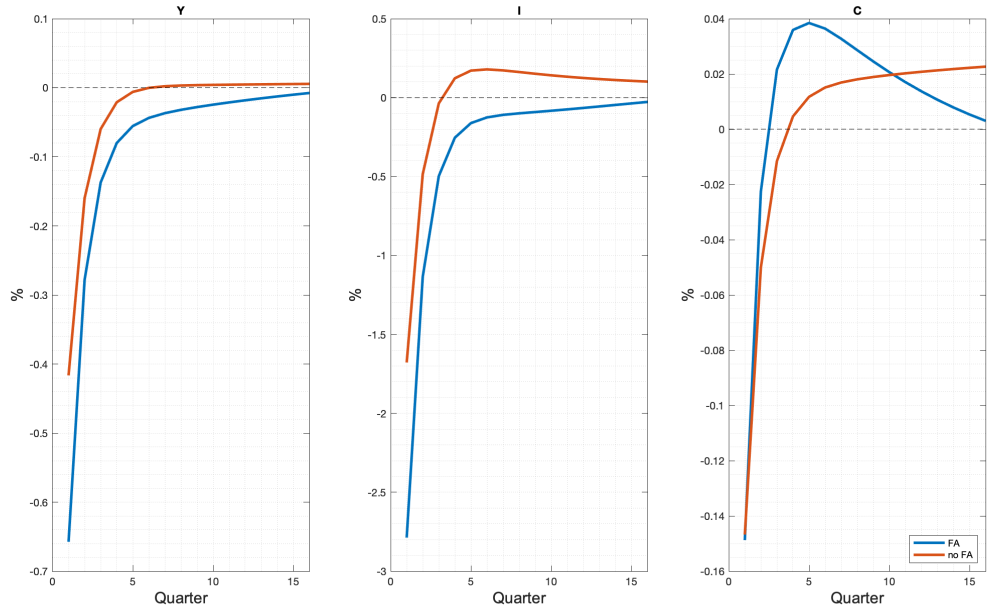


Figure 3: Impulse response to a monetary contraction for aggregate variables
Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

5.2 Inequality among households: wealth

Having agents with different levels of wealth and productivity, monetary shocks alter how households save and consume. The presence of financial frictions in the economy boosts this dynamic. To see this, I start by showing how wealth is affected by an increase in the nominal interest rate through one of the most famous and used indicators of inequality, the Gini index. Figure 4 displays how the Gini index changes from its steady-state value, with or without financial frictions in place.

Regardless of the presence of a financial accelerator, the increase in the Gini index for wealth is significant and long-lasting over time. After hitting its apex one year after the shock, it reverts very slowly. After five years, the increase from steady-state values is still greater than the on-impact fluctuation value. This result suggests that, unlike the aggregate variables analyzed in Figure 3, which revert in a relatively short time, monetary policy can actually have persistent effects on the distribution of wealth among households.[17]

To answer the initial question posed in this paper, Figure 4 tells us that the financial accelerator is, in fact, an inequality accelerator. The green dotted line represents the percentage variation of the wealth Gini index impulse response from the scenario with

[17]It is worth noting that in my model, saying "wealth" is equal to say "liquidity", since bonds and deposits are the only assets through which households can accumulate resources. However, considering multiple assets should not significantly change this dynamic. For instance, Luetticke (2021) considers a contractionary MP shock in a model with liquid and illiquid assets, and the evolution of the Gini index for wealth (see Figure 1 in its Appendix) is similar to that of my model.

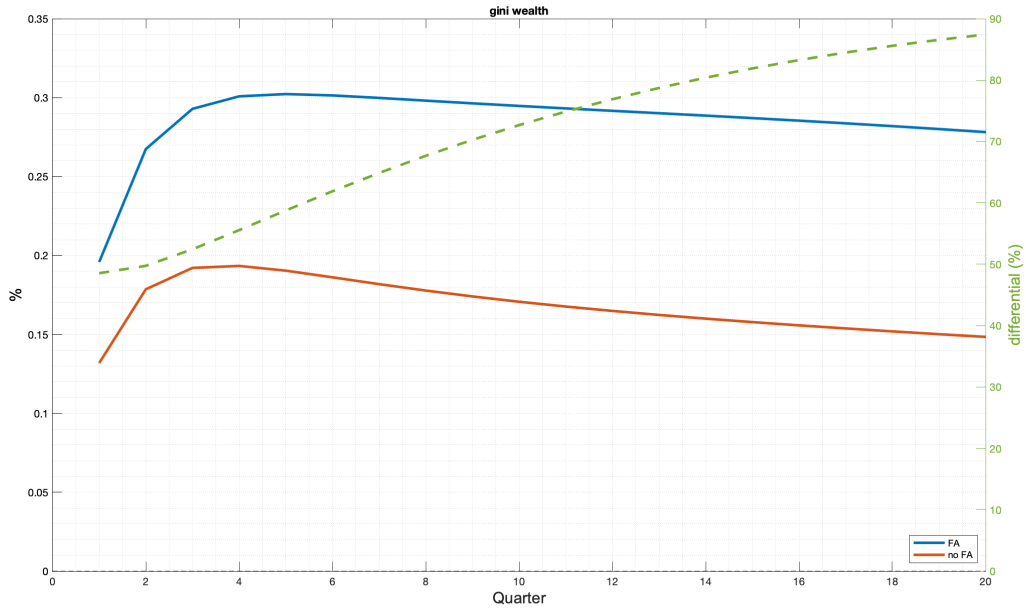


Figure 4: Impulse responses of the Gini index for wealth

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

no financial accelerator to that where frictions are present. On impact, financial frictions imply a wealth inequality fluctuation that is approximately 50% higher. Although both curves (blue and red solid lines) start reverting to the steady-state value after one year, they do so at different paces. The shape of the green-dotted line shows how this difference actually keeps growing in time, reaching almost 90% after 20 quarters. Therefore, not only does the financial accelerator raise wealth inequality, but this increase is constant in the medium-long run. Interestingly, enhancing the magnitude of the monetary shock indeed has consequences on absolute terms (the Gini index increases more with respect to its steady-state values for both scenarios), but not on relative ones, that is, the green-dotted line has almost the same shape and magnitude. The inequality acceleration has little to do with the shock magnitude, while it depends mostly on steady-state dynamics such as leverage and initial wealth distribution. For instance, Figure D.1 in Appendix D shows how the same aggregate shock applied to a similar model that features higher firm leverage (2.5, instead of 2) generates significantly higher Gini index differentials between the case of active and inactive financial frictions.

However, the Gini index cannot reveal how wealth is scattered among agents. To shed some light on this point, I compute the fluctuation of two measures: the share of households perfectly constrained (i.e., with zero wealth) and the share of wealth held by the richest 10 % of the population. The results are presented in Figure 5.

Both measures increase after a rise in the nominal interest rate, and their increase is hump-shaped, as in the case of the Gini coefficient. This suggests: (i) an increase

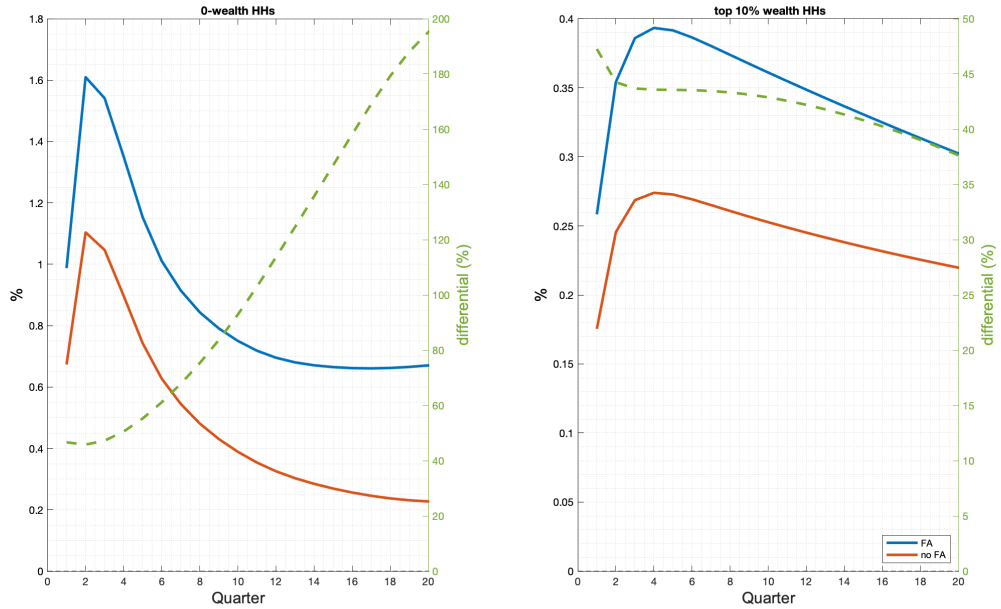


Figure 5: Impulse responses for households' share measures

The graph on the left-hand side represents the fluctuations in the share of households with zero wealth. The one on the right-hand side represents the fluctuations in the share of wealth held by the richer 10%.
Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

in the number of poorer households, since more are now pushed to the constraint, (ii) rich households are getting richer thanks to their gains from interest rates and, if they are rentiers, profits. Therefore, the increase in the Gini index is caused by substantial movements on both tails of the wealth distribution. Similar to Figure 4, the financial accelerator persistently increases the magnitude of the impulse responses. However, the shape of this persistence changes depending on the measure that is at stake. The financial accelerator increases the share of wealth of the richest 10% by 45% on impact, but this acceleration seems to decrease over time in a downward path. On the other hand, the spread between the two scenarios for the zero-wealth household share has an upward path, starting from a similar percentage on impact and reaching almost 200% after 20 quarters. Therefore, while the financial accelerator increases wealth inequality at a general level, it does so differently across the distribution. Specifically, the effects on poorer households seem to be longer-lasting. This result is in line with the initial intuition of the study, where poorer households are more affected by this financial friction because the latter directly affects their main source of income. Richer households are affected significantly as well, mostly due to the fact that wealth in this environment is only liquid and, hence, highly sensitive to changes in the interest rate. Nonetheless, the financial accelerator seems to have less persistent effects on them over time.

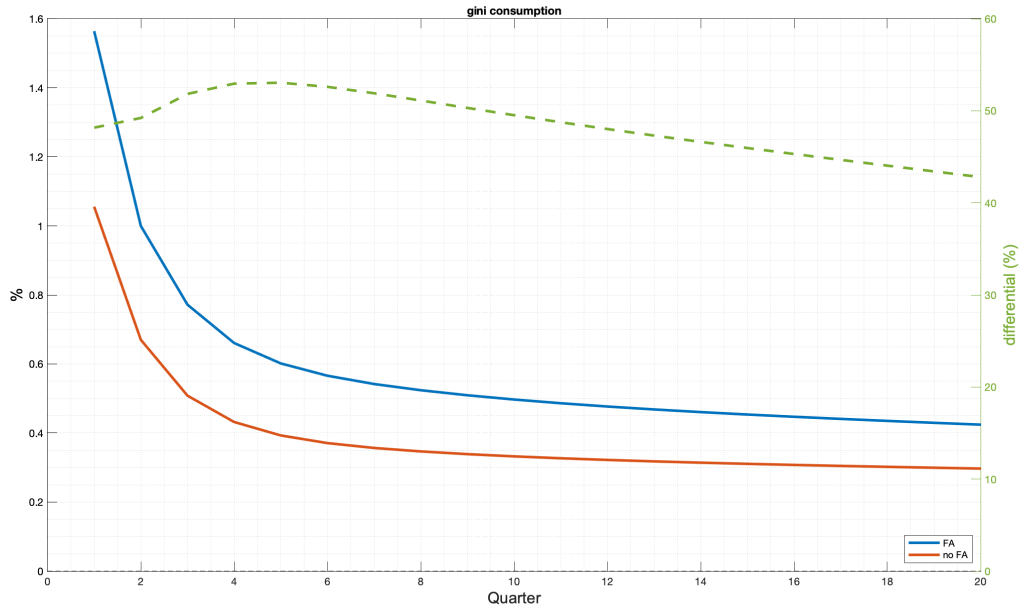


Figure 6: Impulse responses of the Gini index for consumption

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

5.3 Inequality among households: consumption

The financial accelerator also has relevant consequences for how household consumption evolves. Using the Gini coefficient again, I show in Figure 6 how the consumption dispersion rises after a contractionary monetary shock.

On impact, the increase in the Gini index for consumption due to friction is similar to that found in the wealth analysis, at approximately 45-50%. Nevertheless, there are consistent differences. First, the impulse responses are not hump shaped. They seem to revert quite rapidly during the first year, although they stabilize and keep reverting towards the steady-state at a very slow pace. This model replicates the high persistence in consumption and wealth inequality generated by an aggregate shock, a common feature in the HANK literature. Second, the acceleration in consumption inequality (represented by the green dotted line) is actually hump-shaped, going downward after one year. This means that the two curves representing consumption dispersion with and without financial frictions show some convergence already in the medium term, although their reversion to the steady-state is far slower.

Again, the Gini index indicates whether there is an increase in consumption inequality, but does not provide information on who consumes less or more after the shock. In Figure 7, I decompose the aggregate impulse response for consumption to determine how it varies along the distribution. I keep track of fluctuations in average consumption for the bottom 20%, 30%, and 50%, and for the top 20% and 10% of households according

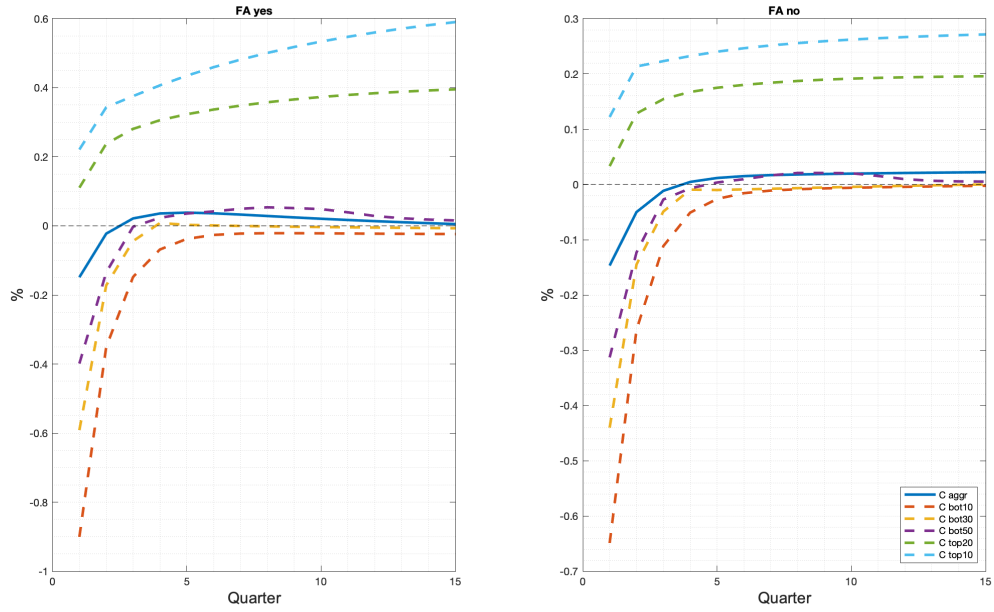


Figure 7: IRFs for consumption, aggregate and averages per wealth share

The blue solid line represents the aggregate consumption. The dashed lines represent the average consumption of a specific share of households.

Monetary shock $\epsilon^R = 0.0009$. On the left is the case with active financial frictions. On the right, financial frictions are shut down.

to their wealth distribution, to study the consumption behavior among poorer and richer agents.[18]

Figure 7 suggests two major considerations. First, monetary contraction results in lower average consumption for poorer households. The average consumption for the bottom 10% of the distribution exhibits a remarkable contraction compared to aggregate consumption. Since these households are perfectly constrained, the reduction in consumption is exclusively due to worsening labor conditions. Consumption depression occurs up to the bottom 50% of the distribution. However, in this last case, consumption overshoots after approximately one year because the bottom half of the population also includes households with a certain level of liquidity, thus benefiting from financial income. On the other hand, the average consumption at the top of the distribution steadily increases when considering both the top 20% and top 10% of the population. Since in this model households can only save in liquid assets, the rise in the interest rate is extremely beneficial for richer households who held a considerable amount of liquidity.

Second, the dynamics presented above are common to the two scenarios considered thus far but with different magnitudes. Although the aggregate consumption path is similar in the two cases, fluctuations in average consumption per wealth share are enhanced in the case of an active financial accelerator. The higher depression in labor and wages

[18]Higher household share values include consumption of lower shares. For example, the average consumption of the bottom 50% also includes consumption of the bottom 30%, and so on.

consequent to active financial frictions is more significant for poorer households, whereas rentiers (many of which are wealthy) benefit from the relative increase in profits. This explains why the Gini index for consumption is higher when we consider the financial accelerator, even though consumption fluctuations appear similar at the aggregate level.

A final remark on consumption inequality concerns the hump-shaped evolution of the percentage difference (green dotted line) between the two Gini coefficients in Figure 6. A possible explanation can be found in the Figure 7. Focusing on the left-hand graph, the average consumption fluctuation for the bottom 50% (purple dotted line) overshoots earlier and is stronger compared with the case without a financial accelerator. Considering that this share includes the bottom 10% (whose average consumption shows a higher and more persistent depression), households around the middle percentile should be responsible for this overshoot. Higher consumption in the middle of the distribution contributes to pushing down the Gini index, causing the reversion of the trend for the green dotted line in Figure 6.

5.4 Inequality between households: skilled-unskilled workers and rentiers

Households in this model are heterogeneous according to their wealth level and their taxed-income source. Consequently, an interesting analysis can be conducted on how inequality is shaped between household types, that is, workers (who collect income through labor) and rentiers(whose income is made of firms' profits). Workers can be further divided into two categories: those with low and high productivity. Since wages are equal for everyone, if two workers with different productivity levels provide the same hours of labor, the high-productivity worker has a higher salary, according to (4). With an abuse of notation, I sort households into three types: unskilled (low-productivity workers), skilled (high-productivity workers), and rentiers (profit collectors). I show how wealth inequality evolves between these households and how the financial accelerator affects these dynamics. To do so, I use the Gini index for wealth for every household type. The results are presented in Figure 8. The financial accelerator not only increases the magnitude of the Gini index fluctuations but, in some cases, also changes the shape of the curves over time. For instance, while unskilled workers' Gini index variation seems to stabilize after one year when financial frictions are shut down, it continues to increase in the other scenario. Variations in the Gini index for rentiers seem to be exactly the same in the two cases, with the difference that they are enhanced in the presence of financial frictions.

Aside from financial accelerator effects, Figure 8 provides two interesting results. First, the group of households with the highest increase in inequality are skilled workers, with changes in the Gini index more than double that of unskilled workers. Second, wealth inequality does not always increase. In fact, wealth inequality decreases among rentiers.

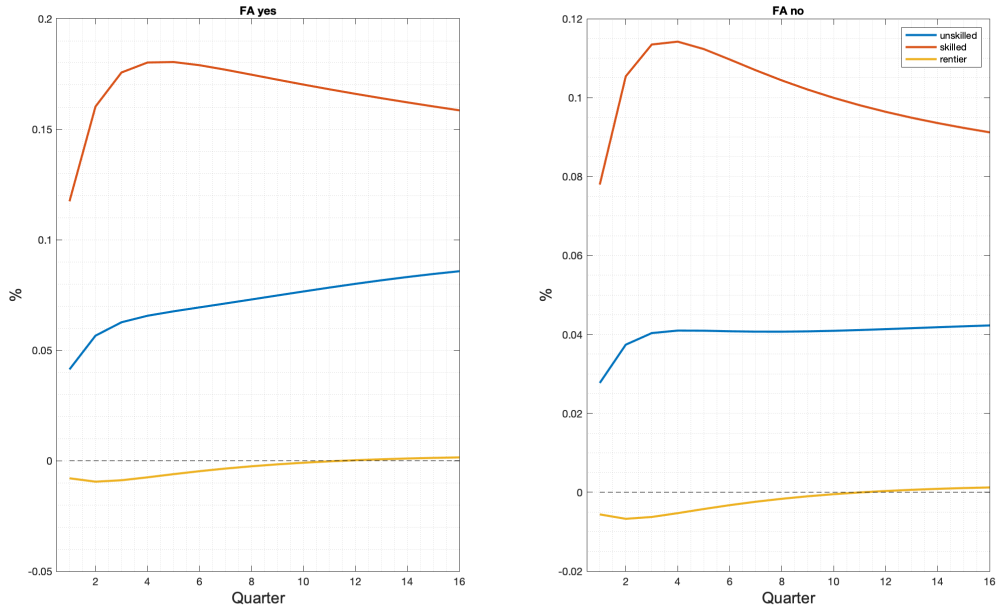


Figure 8: Gini index for wealth inequality according to households type

These trends could be caused not only by households moving along the distribution but also by wealth movements between household types. To see this, I plot in Figure 9 how relative wealth changes after the contractionary monetary shock between workers and rentiers.[19]

Fluctuations in the two scenarios have essentially the same shape, but differ in magnitude. The relative wealth of rentiers increases with a peak of over 3% in the case of financial frictions, whereas the relative wealth of workers decreases, albeit to a lesser extent. The workers who experienced the highest fluctuations are skilled workers who lose more relative wealth than unskilled workers. These results indicate that, aside from changes in household wealth distribution, the variations in the Gini index per household type could also be caused by wealth movements between groups, with the relative gains of rentiers more evenly distributed among them.

5.5 Inequality among households: empirical evidence

In this section, I use empirical data from time series on household consumption and bond premium to provide evidence of the relationship between the EFP and consump-

[19] As "relative wealth" I mean the percentage of wealth in the hands of a certain household type over the whole wealth. Intuitively, if the relative wealth of a household category decreases, this does not necessarily mean that they have less wealth in absolute terms. In fact, as household savings increase after a rise in the nominal interest rate, the opposite is more likely. However, a decrease in relative wealth means a decrease in the weight of a household type's wealth compared to total wealth in the economy. This can be thought as a "relative drain" of wealth from certain households categories at the expenses of others.

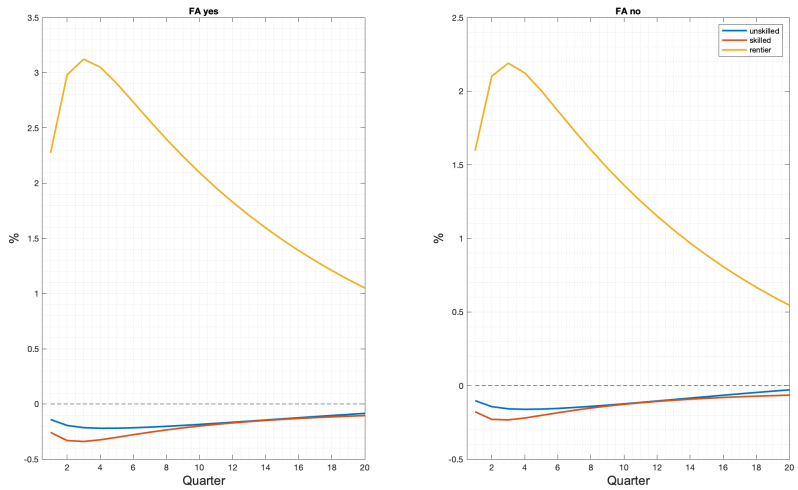


Figure 9: Relative wealth changes per households type

tion inequality. While in my model I study two different types of inequalities (wealth and consumption), doing wealth empirical analysis is less feasible, since the Survey of Consumer Finances (the most reliable source for wealth statistics at the household level in the US) is a triennial statistical survey. For this reason, I consider only consumption inequality, given that the time series of household consumption in the US can be found with quarterly data. Consumption inequality measures are a standard feature of the empirical literature. For instance, Coibion et al. (2017) use consumption data from the Consumer Expenditure Survey (CEX) to study the effects of monetary policy shocks on, among others, consumption dispersion (both as ratios of consumption percentiles and Gini index) and find that contractionary monetary shocks increase this measure. In light of these factors, I also use data from the CEX to build the inequality measure.

Regarding the EFP, different interest rate spreads can be used as a proxy. For instance, Bernanke et al. (1999) use in their empirical analysis two different spreads to represent the EFP: the difference between the six-month commercial paper rate and the six-month T-bill rate and the difference between the prime lending rate and the six-month T-bill rate. For my analysis, however, I choose to employ the "GZ credit spread" built by Gilchrist and Zakrajšek (2012) as a proxy for the spread between non-financial firms and risk-free assets.

Similarly to Lee et al. (2020), I employ local projections á la Jordà (2005) to determine whether an increase in the EFP is related to an increase in consumption dispersion:

$$c_{t+h}^{disp} = \beta_h^1 + \beta_h^2 s_t^{GZ} + \beta_h^3 X_{t-1} + \epsilon_{t+h} \ , \tag{36}$$

where $h \geq 0$ is the forecast horizon set to 16 quarters, c^{disp} is the measure of consumption

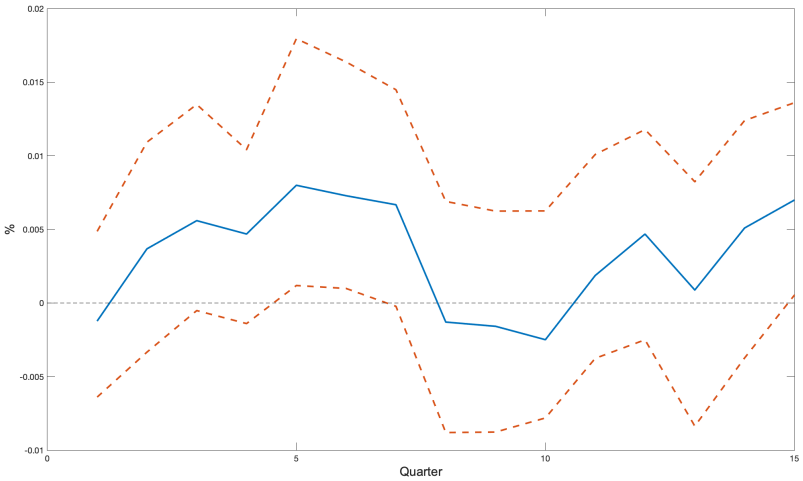


Figure 10: IRFs of consumption dispersion to the credit spread

IRFs computed through local projections. The blue solid line represents point estimates. The red dotted lines delimit the 90% confidence interval calculated with a block bootstrap.

dispersion,[20] s^{GZ} is the GZ credit spread that proxy the EFP and X a vector of controls that includes a one-period lag of consumption dispersion, credit spread, inflation, GDP, real investment, labor productivity and charge-off rate on business loans.[21] All these variables are expressed in logs and are quarterly data for the same period investigated in the model part of the paper, that is, 1985q1-2007q4. All data are detrended with a quadratic trend. Furthermore, CEX data are smoothed with a centered three-quarter moving average and subject to a seasonal dummy. Appendix I contains more information about the data used in these regressions.

The results are shown in Figure 10. An increase in the proxy for the EFP is connected to a rise in consumption inequality, validating the results of the model. Except for an initial decline for the first period in point estimates (which is not significant at the 90% confidence interval), local projection values are positive most of the time, being statistically significant after one year for three quarters.

A further empirical analysis I conduce is to change the independent variable, the credit spread, with leverage values.[22] In Appendix I, equation (A8) states that higher spreads are linked to a higher level of leverage. If this result also holds in the empirical data, I should see a positive relationship between consumption dispersion and leverage as well. Therefore, I rewrite the local projection as:

[20] I measure consumption dispersion as the ratio between the consumption at the 50th percentile over the consumption at the 10th percentile of the consumption distribution, a common index in the literature.

[21] As emphasized by Lee et al. (2020), this last variable is particularly relevant because it controls for the importance of business defaults on the credit spread.

[22] I only consider leverage ratios for non-financial corporate firms.

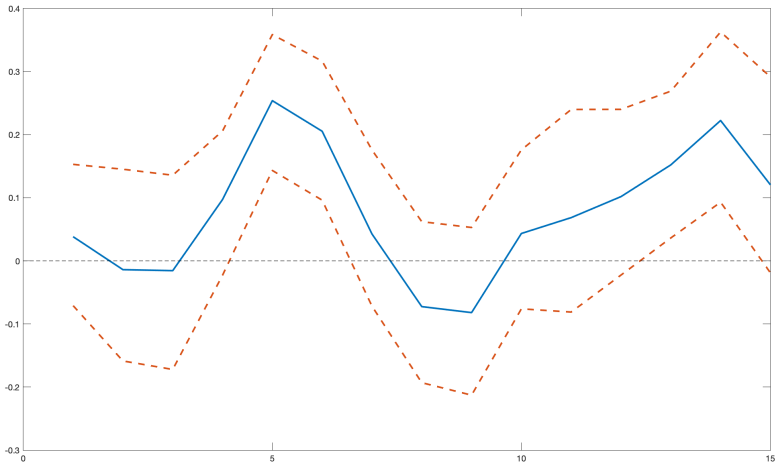


Figure 11: IRFs of consumption dispersion to the leverage

IRFs computed through local projections. The blue solid line represents point estimates. The red dotted lines delimit the 90% confidence interval calculated with a block bootstrap.

$$c_{t+h}^{disp} = \beta_h^1 + \beta_h^2 LEV_t + \beta_h^3 X_{t-1} + \epsilon_{t+h} \ , \tag{37}$$

and display results in Figure 11. The leverage time series is detrended with a quadratic trend.

When compared to the results with credit spread as the dependent variable, the point estimates show stronger fluctuations, but the path seems reasonably similar. It is then safe to assume that local projection results show that an increase in firms' leverage is indeed connected with an increase in consumption dispersion among households.

6 Concluding remarks

Introducing financial frictions in the flavor of Bernanke et al. (1999) in a full-fledged HANK model, I show that the financial accelerator has important consequences at both aggregate and idiosyncratic levels. In the face of a contractionary monetary policy shock, the financial accelerator not only causes a higher depression in aggregate variables such as output and investment, but also an increase in inequality measures concerning wealth and consumption, coming to the conclusion that the financial accelerator is also an "inequality accelerator".

This acceleration is mostly due to movements at the two tails of the wealth distribution, with constrained households playing a crucial role. Since they cannot rely on either savings or borrowings,to smooth consumption, they rely only on their income, which is for the greater part coming from labor. Frictions on the production side of the economy, as the ones introduced in this paper, depress labor income, pushing more people to the

borrowing constraint and increasing wealth and consumption inequality.

Financial frictions not only enhance wealth changes among households but also between household types (workers and rentiers). In terms of relative wealth shares in the hands of a certain group, rentiers become relatively richer, and workers become relatively poorer after an increase in the interest rate. Wealth inequality in the economy is not only a matter of how households move along the wealth distribution but also of how wealth is redistributed between household types.

I also provide empirical evidence to study the relationship between consumption dispersion and the corporate credit spread. This spread is a good proxy for the EFP built into the theoretical model, which in turn is an indicator of how strong financial frictions are (the greater the frictions, the greater the EFP). The results from local projections show that an increase in the credit spread is associated with a rise in consumption dispersion, qualitatively validating the dynamics of the model. The results hold also when I consider leverage values for corporate firms instead of the corporate spread.

Although central bankers do not formally care about redistribution trends, their concern about this topic has increased over the last decade. From a technical perspective, the blooming literature on HANK models proves that wealth distribution has important effects on the transmission of monetary policies. Knowing that the financing structure of non-financial firms has important implications for the wealth and consumption redistribution of monetary policy shocks could be something to consider for future policy-making.

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Appendix

A Idiosyncratic productivity process and the joint distribution

Households can be workers, with productivity $h > 0$, or rentiers, with $h = 0$, which means that they do not earn labor income but only profit income. Furthermore, I assume that there are only two possible productivity realizations for workers: high productivity, h^H, and low productivity, h^L. This assumption, in addition to simplifying the computations, is useful for developing the analysis in Section 5.4 between skilled and unskilled workers. The Markov process generates the following transition matrix:

$$
\begin{array}{cc}
 & h_{t+1} \\
 & \begin{array}{ccc} h^L & h^H & 0 \end{array} \\
h_t \begin{array}{c} h^L \\ h^H \\ 0 \end{array} & \begin{bmatrix} p_{LL}(1-\zeta) & p_{LH}(1-\zeta) & \zeta \\ p_{HL}(1-\zeta) & p_{HH}(1-\zeta) & \zeta \\ 0 & \iota & 1-\iota \end{bmatrix}
\end{array}
$$

with probabilities, p, determined using the Tauchen method. In other studies using this household distribution framework, such as Luetticke (2021),rentiers who become workers are endowed with the median productivity level ($h = 1$). However, in this model, there are no states with median productivity levels.[23] Therefore, I assume that new workers are endowed with the highest productivity possible, h^H.

At the steady state, a joint distribution of households exists according to their wealth level, a, and their productivity, h. This joint distribution can be represented by the bi-dimensional matrix as follows:

$$
\begin{array}{cc}
 & \begin{array}{c} h_m \\ \cdots \\ h_2 \\ h_1 \end{array} \\
\text{prod. } h & \begin{bmatrix} H_{m,1} & H_{m,2} & \cdots & H_{m,n} \\ \cdots & \cdots & \cdots & \cdots \\ H_{2,1} & H_{2,2} & \cdots & H_{2,n} \\ H_{1,1} & H_{1,2} & \cdots & H_{1,n} \end{bmatrix} \\
 & \begin{array}{cccc} a_1 & a_2 & \cdots & a_n \end{array} \\
 & \text{wealth } a
\end{array}
$$

where $H_{1,1}$ is the share of households with the lowest level of wealth and labor productivity (except for the last state $h_m = 0$, since in this model they are rentiers), and $\int H dadh = 1$.

[23] Following the calibration of the baseline model, I obtain that $h^L = 0.786$ and $h^H = 1.272$

As the vector indicating possible household wealth levels is composed of 100 entries, this joint distribution matrix comprises 300 grid points ($a_n = 100$ and $h_m = 3$).

B Entrepreneurs optimal contract

B.1 Idiosyncratic shock on return on capital

I assume that the Idiosyncratic shock ω is distributed log-normally. i.e. $\omega \in [0, +\infty)$.[24] Using results from Appendix A.2 in Bernanke et al. (1999) I can write $F(\omega)$, $\Gamma(\omega)$ and $G(\omega)$ in the analytical expressions that I use in my code to solve the model:

$$F(\omega) = \Phi\left[\left(log(\bar{\omega}) + \frac{1}{2}\sigma_\omega^2\right)/\sigma_\omega\right] , \tag{A1}$$

$$\Gamma(\omega) = \Phi\left[\left(log(\bar{\omega}) - \frac{1}{2}\sigma_\omega^2\right)/\sigma_\omega\right] + \bar{\omega}\left\{1 - \Phi\left[\left(log(\bar{\omega}) + \frac{1}{2}\sigma_\omega^2\right)/\sigma_\omega\right]\right\} , \tag{A2}$$

$$G(\omega) = \Phi\left[\left(log(\bar{\omega}) + \frac{1}{2}\sigma_\omega^2\right)/\sigma_\omega - \sigma_\omega\right] . \tag{A3}$$

With $\Phi(\cdot)$ being the normal cumulative distribution function and σ_ω the standard deviation of the idiosyncratic shock on entrepreneurs' return on capital.

B.2 Financial intermediaries' participation constraint and entrepreneur j's optimization problem

After substituting (9) and (8) into (10), I obtain:

$$[1-F(\bar{\omega}_{jt+1})]\bar{\omega}_{jt+1}R^K_{t+1}q_tK_{jt+1}+(1-\mu)\int_0^{\bar{\omega}_{jt+1}}\omega_j dF(\omega_j)R^K_{t+1}q_tK_{jt+1} = \frac{R_{t+1}}{\pi_{t+1}}(q_tK_{jt+1}-N_{jt+1}) . \tag{A4}$$

Divide everything by $R^R_{t+1}q_tK_{jt+1}$:

$$\frac{R^K_{t+1}}{\frac{R_{t+1}}{\pi_{t+1}}}\left([1 - F(\bar{\omega}_{jt+1})]\bar{\omega}_{jt+1} + (1-\mu)\int_0^{\bar{\omega}_{jt+1}}\omega_j dF(\omega_j)\right) = \left(1 - \frac{N_{jt+1}}{q_tK_{jt+1}}\right) . \tag{A5}$$

Following the notation used in Bernanke et al. (1999) and Christiano et al. (2014):

[24] Note that other kinds of distribution with values greater or equal to 0 could be used as well. Here I choose to adapt the same distribution as in Bernanke et al. (1999) to give a sense of continuity between the two studies.

$$\Gamma(\bar{\omega}_j) \equiv \int_0^{\bar{\omega}_j} \omega_j dF(\omega_j) + \bar{\omega}_j \int_{\bar{\omega}_j}^{\infty} dF(\omega_j) \ , \qquad \mu G(\bar{\omega}_j) \equiv \mu \int_0^{\bar{\omega}_j} \omega_j dF(\omega_j) \ , \tag{A6}$$

where $\Gamma(\bar{\omega}_j)$ is the expected gross share of profits going to the lender and $\mu G(\bar{\omega}_j)$ is the expected monitoring cost paid by the lender. $\Gamma(\bar{\omega}_j)$ can be rewritten as:

$$\Gamma(\bar{\omega}_j) = G(\bar{\omega}_j) + \bar{\omega}_j \left[1 - F(\bar{\omega}_j)\right] \ . \tag{A7}$$

I can now use (A6) and (A7) in (A5) and rearrange to finally obtain:

$$\frac{R^K_{t+1}}{\left(\frac{R_{t+1}}{\pi_{t+1}}\right)} = \frac{1}{\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})} \left(1 - \frac{N_{jt+1}}{q_t K_{jt+1}}\right) \ , \tag{A8}$$

where $\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})$ is the share of entrepreneur j's profits going to the lender (as loan repayment), net of auditing costs.

Equation (A8) is the complete version of (11), which explain the function underlying $f\left(\bar{\omega}_{jt+1}, LEV_{jt+1}\right)$. For a higher level of entrepreneur leverage, the EFP increases, raising the return on capital. However, it also increases the probability of an entrepreneur's default, thereby increasing the net share of profit demanded by financial intermediaries as loan repayment, resulting in higher financing costs for entrepreneurs. To see in detail how this mechanism works, I show the entrepreneur j's optimization problem below.

According to the optimal contract set by financial intermediaries, entrepreneur j's expected return can be expressed as:

$$E_t \left\{ \int_{\bar{\omega}_{jt+1}}^{\infty} \omega_j dF(\omega_j) R^K_{t+1} q_t K_{jt+1} - (1 - F(\bar{\omega}_j)) R^K_{t+1} q_t K_{jt+1} \right\} \ , \tag{A9}$$

with expectations taken with respect to the realization of R^K_{t+1}. The first term of (A9) represents the entrepreneur's profit when she does not default on debt, while the second term is the amount of profits that she uses to repay the lender. Following the notation used above, and considering that the entrepreneur's return is subject to the participation constraint (10), I write entrepreneur j's optimal contracting problem as:

$$\begin{aligned} &\max_{\{K_{jt+1}, \bar{\omega}_{jt+1}\}} E_t \left\{ \left[1 - \Gamma(\bar{\omega}_{jt+1})\right] R^K_{t+1} q_t K_{jt+1} \right\} \ , \\ &s.t. \quad \frac{R_{t+1}}{\pi_{t+1}} (q_t K_{jt+1} - N_{jt+1}) = \left[\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})\right] R^K_{t+1} q_t K_{jt+1} \ . \end{aligned} \tag{A10}$$

Deriving F.O.C. I obtain:

$$w.r.t.\ \ \omega_{jt+1}: \quad -\Gamma'(\bar{\omega}_{jt+1}) + \lambda_{jt+1}\left[\Gamma'(\bar{\omega}_{jt+1}) - \mu G'(\bar{\omega}_{jt+1})\right] = 0\ , \tag{A11}$$

$$w.r.t.\ \ K_{jt+1}: \quad E_t\left\{\left[1-\Gamma(\bar{\omega}_{jt+1})\right]R^K_{t+1} - \lambda_{jt+1}\left[\frac{R_{t+1}}{\pi_{t+1}} - \left(\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})R^K_{t+1}\right)\right]\right\} = 0\ , \tag{A12}$$

$$w.r.t.\ \ \lambda_{jt+1}: \quad E_t\left\{\frac{R_{t+1}}{\pi_{t+1}}(q_tK_{jt+1} - N_{jt+1}) - \left[\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})\right]R^K_{t+1}q_tK_{jt+1}\right\} = 0\ , \tag{A13}$$

where λ_j is the Lagrangian multiplier for entrepreneur j's problem. By rearranging (A11), it is possible to express λ_{jt+1} as a function of only $\bar{\omega}_{jt+1}$. Furthermore, rearranging (A12):

$$E_t\left\{\frac{R^K_{t+1}}{\frac{R_{t+1}}{\pi_{t+1}}}\right\} = \frac{\lambda_{jt+1}}{\left[1-\Gamma(\bar{\omega}_{jt+1}) + \lambda_{jt+1}\left(\Gamma(\bar{\omega}_{jt+1}) - \mu G(\bar{\omega}_{jt+1})\right)\right]}\ . \tag{A14}$$

It can be proven that there is a monotonically increasing relationship between the EFP and $\bar{\omega}_j$. According to (A8), we can extend this relationship between the EFP and the leverage level of j, assessing that a higher entrepreneur's leverage implies a higher EFP.[25]

Furthermore, it is clear from (A14) that $\bar{\omega}_j$ is determined only by aggregate variables. Thus, any entrepreneur chooses the same threshold $\bar{\omega}$ for the idiosyncratic shock on capital returns, below which they default, and the same leverage level.[26] This result allows to consider only the aggregate variables in the production sector part of the model, since every entrepreneur has the same firm structure.

C Impulse responses of MP contractionary shock

Figure C.1 show several aggregate variables impulse responses for the monetary policy shock considered in the main text. This integrate IRFs for GDP, consumption and investment present in Figure 3 in the main text.

[25] See Appendix A.1 in Bernanke et al. (1999) for proofs.

[26] According to (A8), leverage is a function of the EFP (composed of only aggregate variables) and $\bar{\omega}_j$. If $\bar{\omega}_j$ depends only on aggregate variables (since it is a function of the EFP, according to (A14)), then the same can be said for the leverage.

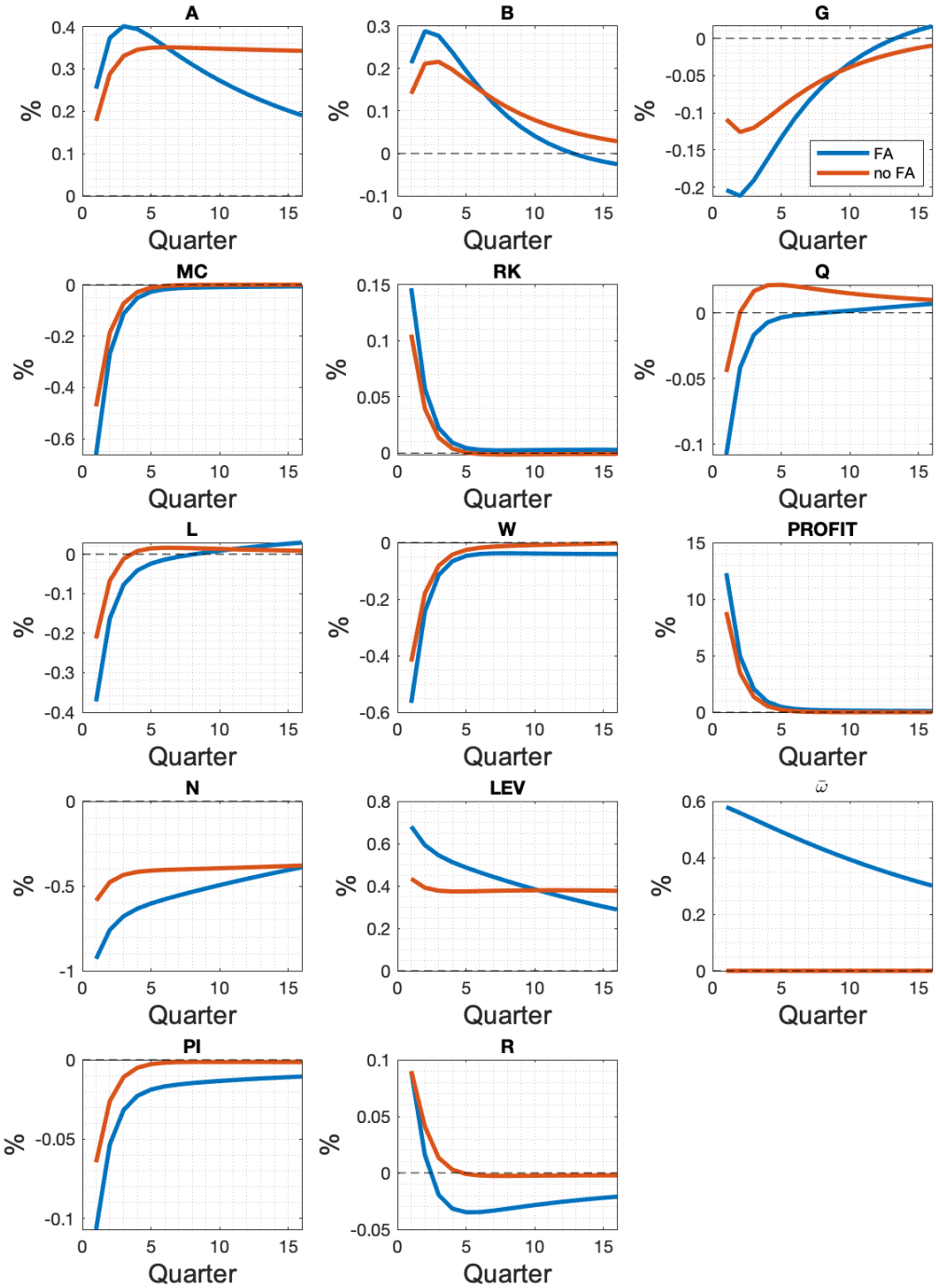

Figure C.1: Aggregate fluctuations consequent to an increase of the nominal interest rate.
Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

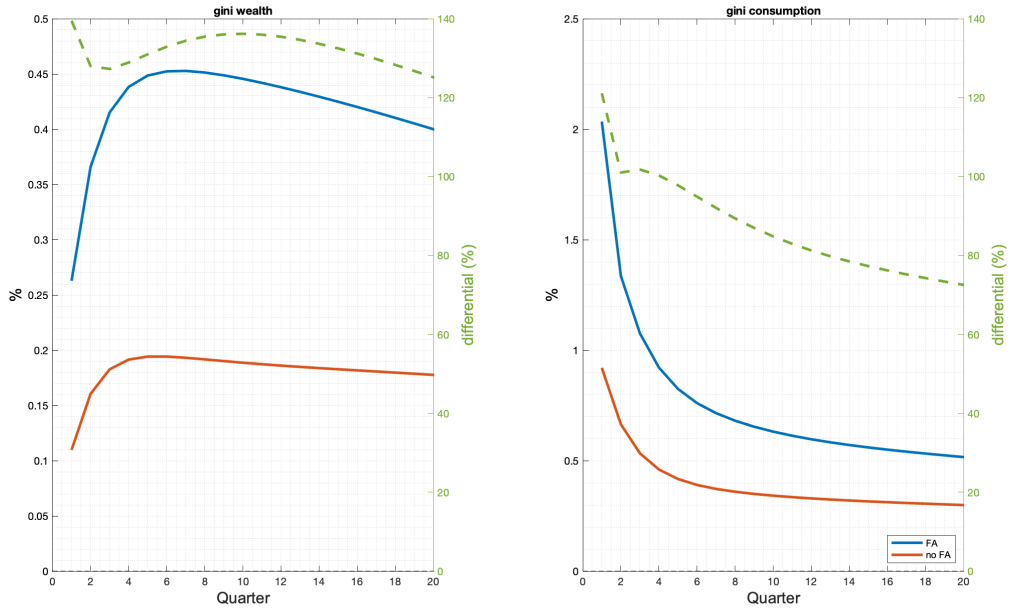

Figure D.1: IRFs for Gini indices, LEV = 2.5

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

D Gini indices for higher leverage at Steady-State

Figure D.1 shows fluctuations of the Gini indices for wealth and consumption in a model with a higher initial level of firm leverage. I show results for the case where the latter is targeted to be equal to 2.5 (instead of 2, as in the baseline model). To reach this level of leverage while maintaining the general calibration, I slightly decrease the discount factor β, increase the labor disutility parameter ψ to 6.5, change the household probability to become a rentier, $\zeta = 0.00056$, and the parameter governing the adjustment cost of capital, $\phi = 10$.

E Robustness to risk aversion

For the baseline model, I used a parameter for households' risk aversion $\xi = 2$, which is already used in other HANK models in the literature. However, other models used different values; for instance, Bayer et al. (2019) and Luetticke (2021) assume $\xi = 4$. I recalibrated the model with this parameter to obtain relevant moments as in the baseline version. This implies a discount factor $\beta = 0.986$, labor disutility parameter $\psi = 11$, household probability of becoming a rentier $\zeta = 0.00072$ and the parameter governing the adjustment cost of capital $\phi = 7$. Figure E.1 and Figure E.2 show fluctuations for aggregate variables and Gini indices, respectively.

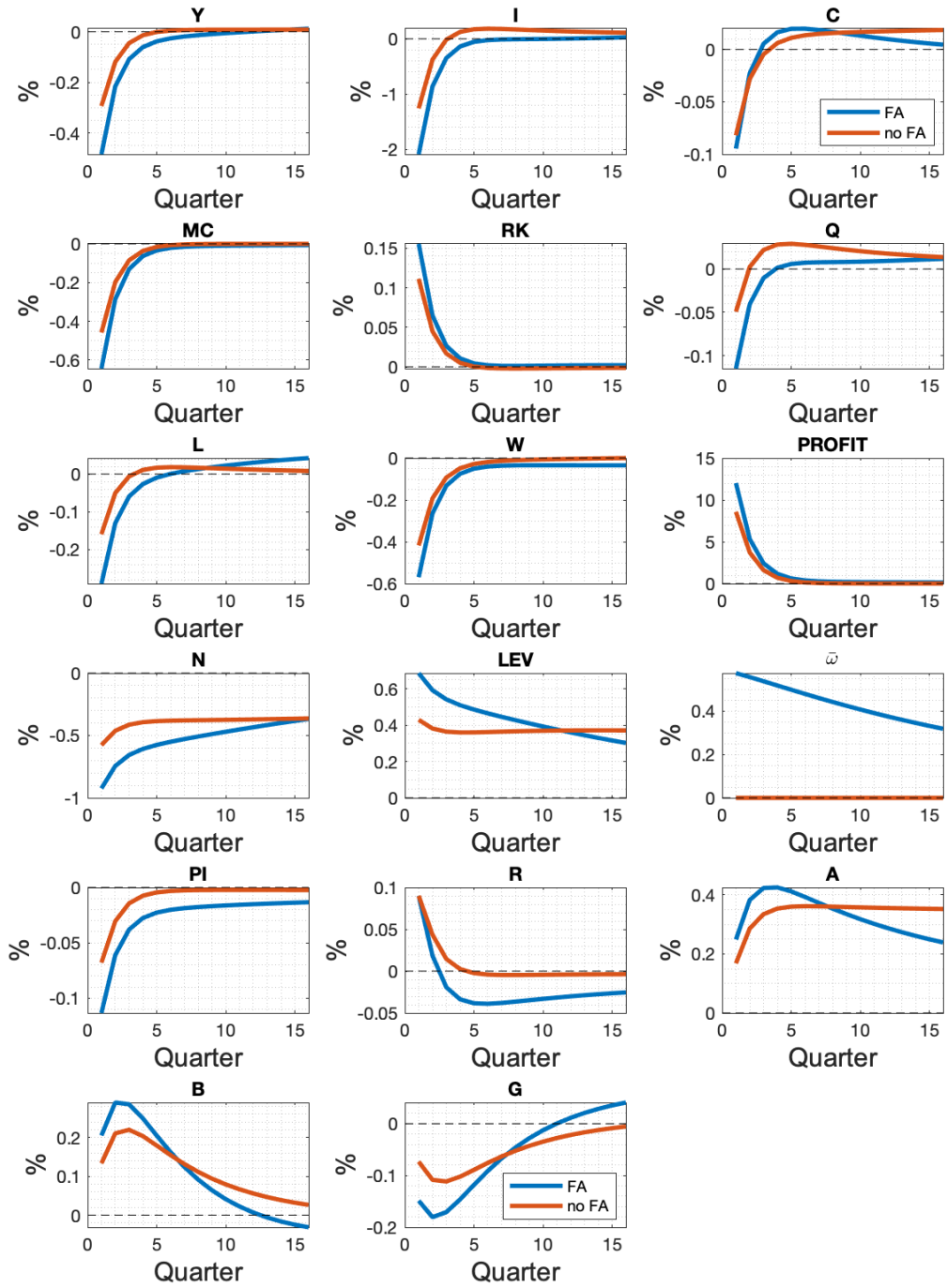

Figure E.1: IRFs for aggregate variables, $\xi = 4$

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

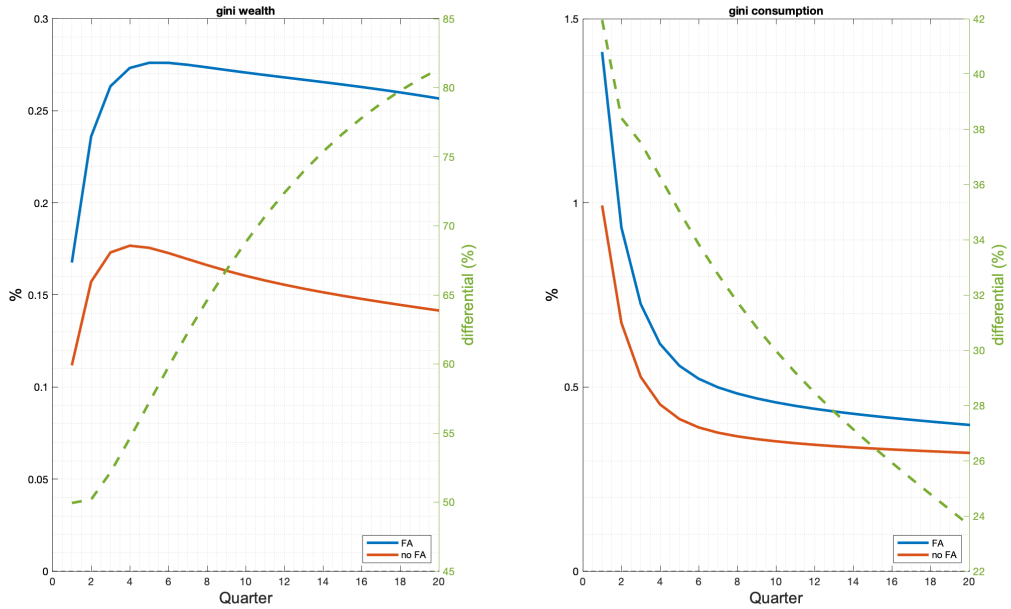

Figure E.2: IRFs for Gini indices, $\xi = 4$

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

F Robustness to investment cost

The baseline model features quadratic investment costs (the central term on the right-hand side of Eq. (22)) where the parameter ϕ is calibrated to match an investment volatility of 4.5. I display in Figure F.1 and Figure F.2 aggregate and Gini index fluctuations for the case limit where there is no investment cost, that is, $\phi = 0$. This means that the capital price q is fixed over time, and entrepreneurs do not make any profit from capital gains or creation of new capital ΔK. This extreme calibration also confirms the financial accelerator: the output, investment, consumption, and Gini indices are all greater when financial frictions are active. However, it is worth noting that some aggregate variables display completely different behaviors. For instance, the quantity of aggregate labor L increases after a MP contractionary shock. Interestingly, aggregate consumption fluctuations do not overshoot in the short-run with this parameterization.

G Robustness to fiscal policy

Since I employ a HANK model, the Ricardian equivalence does not hold, and fiscal policies could have significant effects on monetary transmission. In the baseline model, I assume that the government adjusts its spending to bring debt to steady-state values. In line with the empirical data, as in Bayer et al. (2019) and Luetticke (2021), I set the

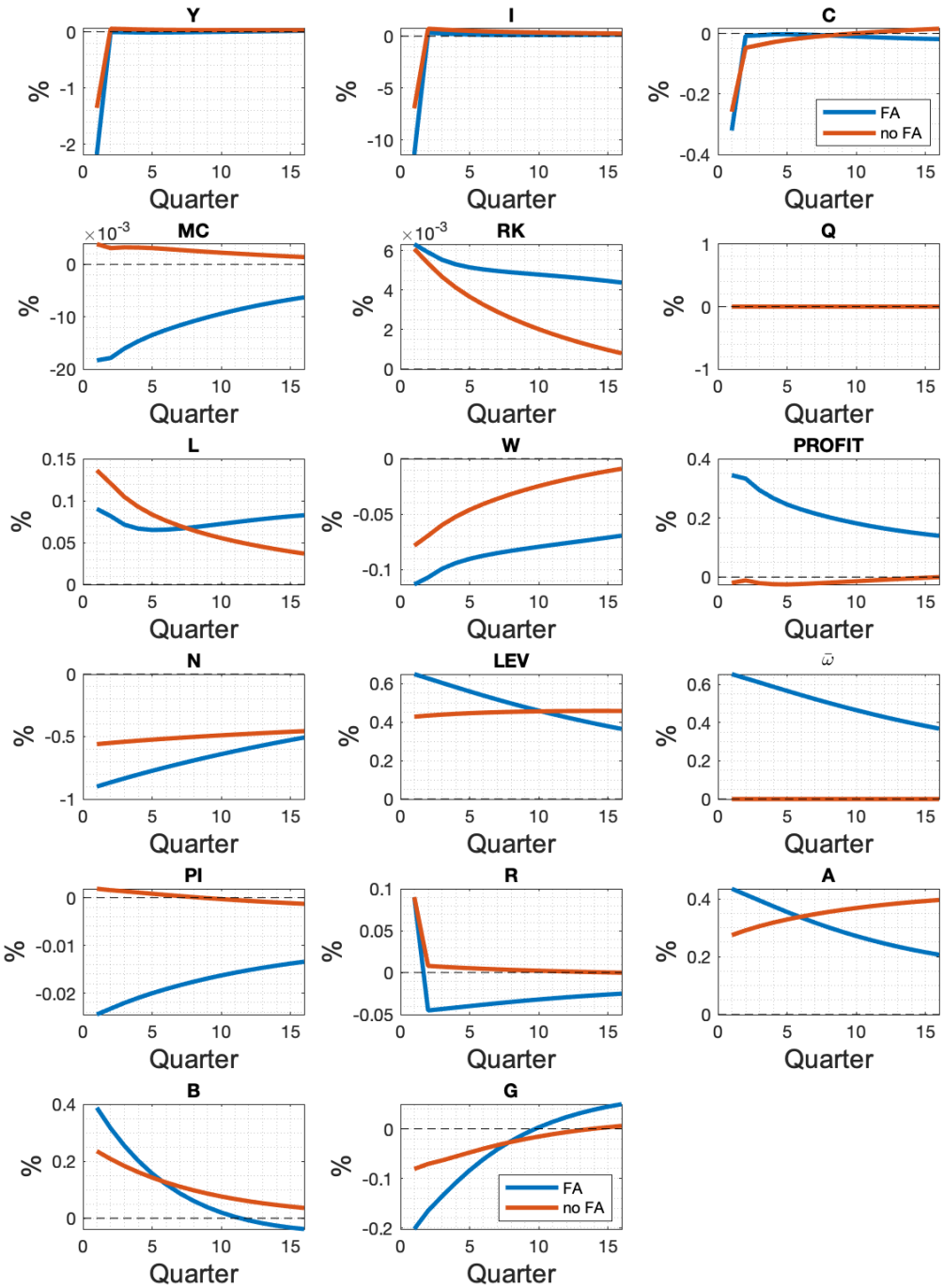

Figure F.1: IRFs for aggregate variables, $\phi = 0$

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

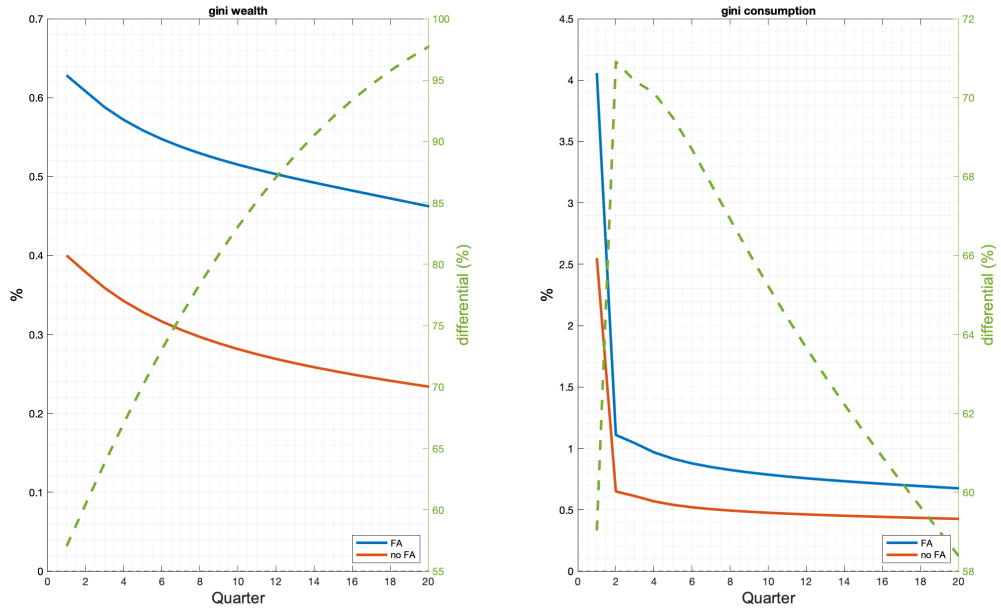

Figure F.2: IRFs for Gini indices, $\phi = 0$

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

debt autocorrelation $\rho_B = 0.86$, meaning that the government is willing to roll over most of this debt, sustaining a higher level of public expenditure. I now consider the case in which the government wants to revert immediately to its steady-state level of debt after a MP contractionary shock, setting $\rho_B = 0$. The results are shown in Figure G.1 and Figure G.2. The government achieves debt control by cutting even more expenditure, inducing a higher economic depression in terms of output and consumption, and increasing inequalities even more when compared to the baseline specification of the model.

The government could also choose to maintain spending at its steady-state level and adjust taxation through the tax parameter τ. The results are shown in Figure G.3 and Figure G.4. While Output and investment do not display significant differences compared to the baseline calibration, consumption falls more, and the respective Gini index is higher. Taxes increase to balance the government budget constraint; however, taxation is proportional and not progressive. Therefore, poorer households (who rely more on labor income for consumption and have a higher marginal propensity to consume) are more affected by this tax rise. In this model, financial income is not taxed; therefore, wealthier households suffer less from an increase in tax rate.

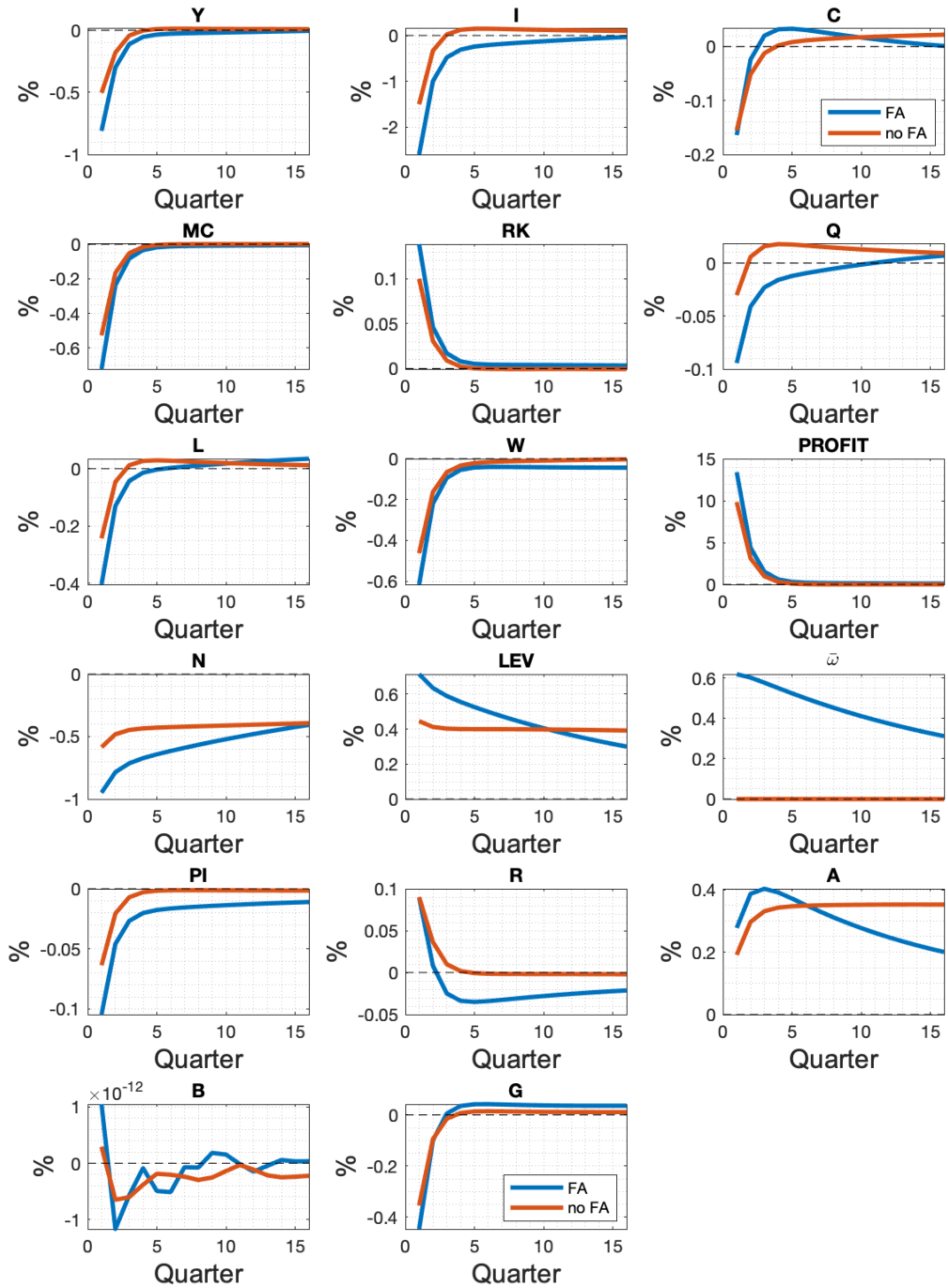

Figure G.1: IRFs for aggregate variables, $\rho_B = 0$

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

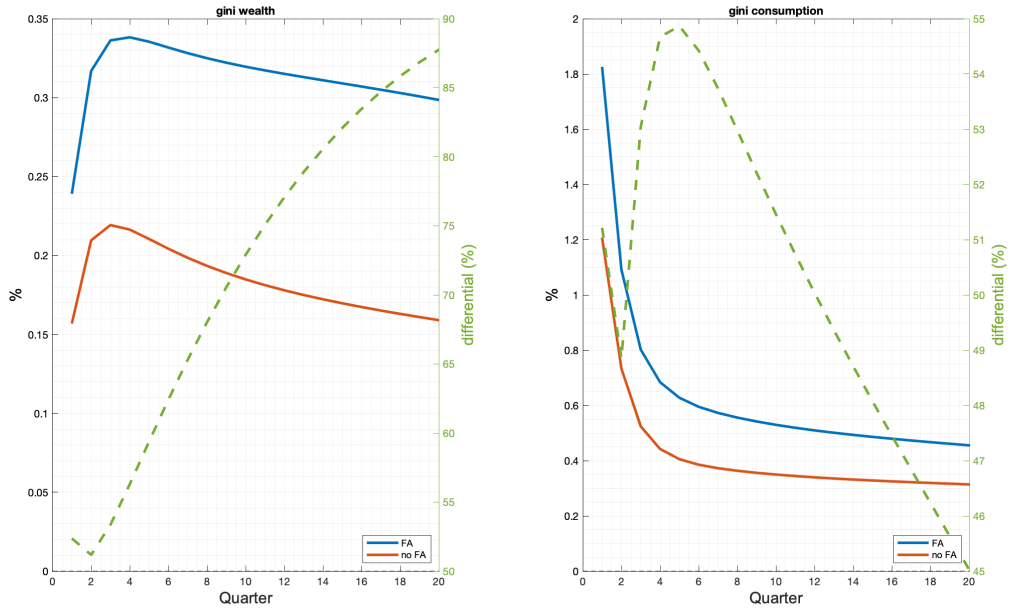

Figure G.2: IRFs for Gini indices, $\rho_B = 0$

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

H Impulse responses of a TFP shock

In this section, I show aggregate and inequality fluctuations when a positive TFP shock occurs, instead of a contractionary MP shock. The shock to z_t follows an AR(1) process with persistence $\rho_z = 0.95$ and SD $\sigma_z = 0.00915$. The SD is calibrated such that the autocorrelation of output is in line with values from US data. Figure H.1 shows aggregate variable fluctuations, while Figure H.2 shows fluctuations of the Gini indices for wealth and consumption. Interestingly, adopting a HANK model seems to solve the "financial accelerator dampening" of the TFP shock that occurs in Bernanke et al. (1999). In their results, the TFP shock confirms the financial accelerator only if persistence is set to $\rho_z = 1$ such that the shock never reverts to zero over time. For a more standard value of the TFP shock persistence, such as $\rho_z = 0.95$, the model presented in Bernanke et al. (1999) shows a "financial deceleration". On the other hand, as shown in Figure H.1, output, investment, and consumption also increase (albeit slightly) when persistence is less than 1.

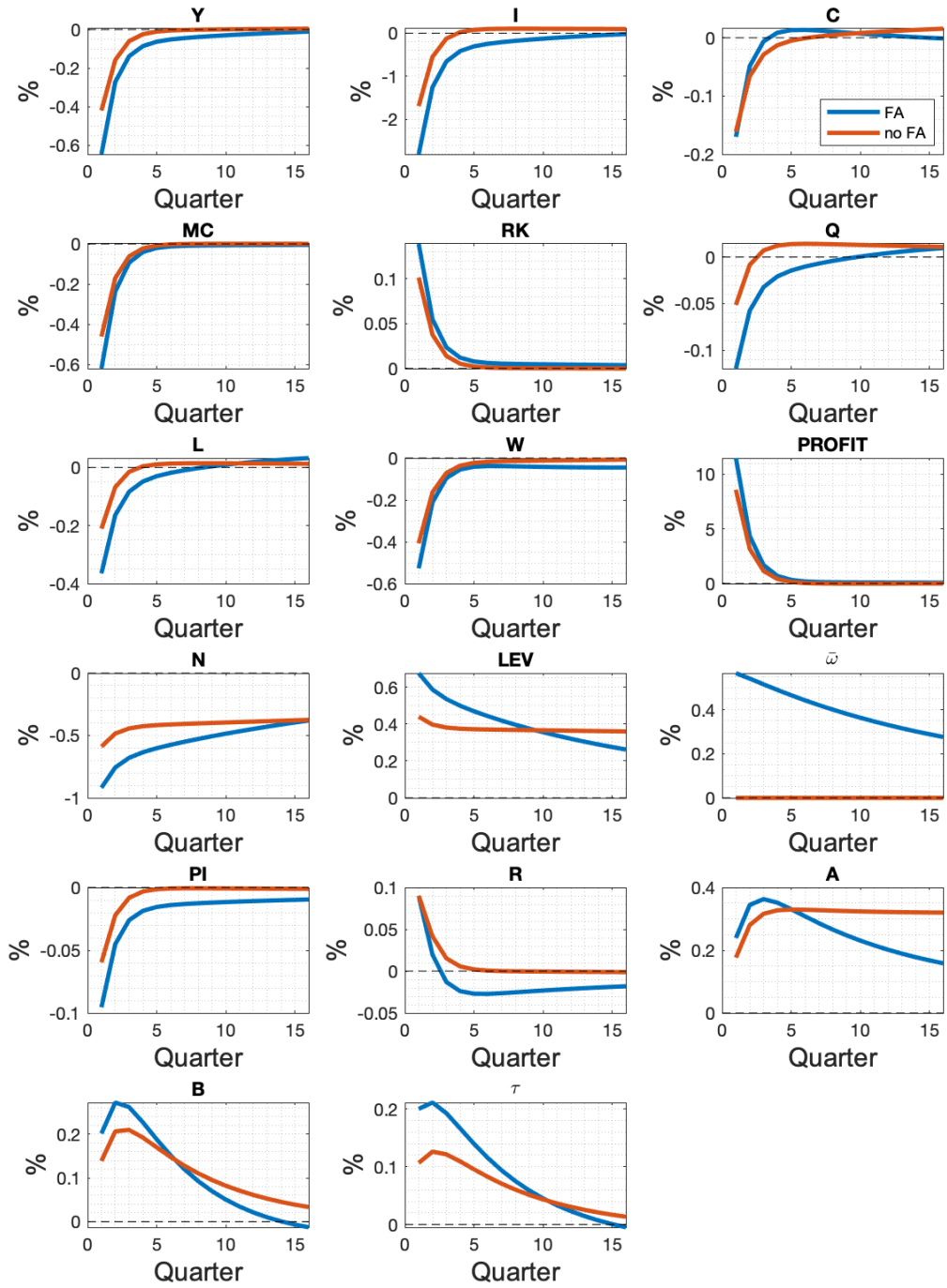

Figure G.3: IRFs for aggregate variables, τ adjustment

Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

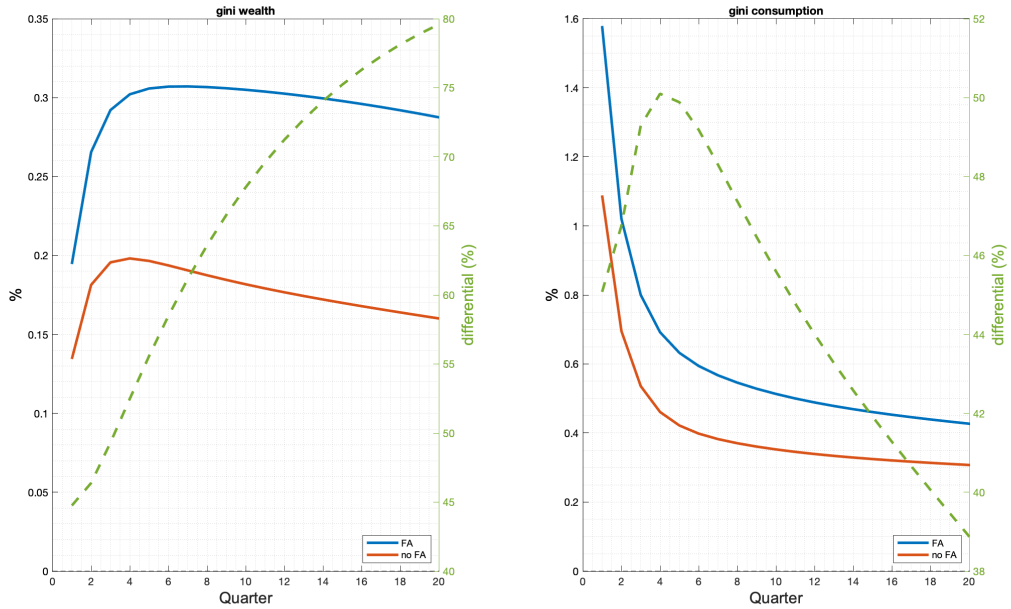

Figure G.4: IRFs for Gini indices, τ adjustment

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. Monetary shock $\epsilon^R = 0.0009$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

I Empirical results data

To build the consumption dispersion measure used as the dependent variable in (36), I use data on consumption expenditure from the Consumer Expenditure Survey (CEX).[27] One of the strengths of this dataset is that it contains consumption expenses at quarterly frequency since the 1980s. Because households in the dataset are representative of a portion of the US population (a weight is assigned to each of them), it is possible to build consumption distributions and consumption inequality measures. As in Lee et al. (2020), I adopt a definition of consumption that includes durables, non-durables, and housing services, and restrict the sample to households with age 20-60, working at least 10 hours a week, with a partner, and families with 10 members or less. Consumption dispersion is represented as the log of the ratio of the consumption level at the 50th percentile to that at the 10th percentile, with percentiles calculated from the consumption distribution. Regarding other variables in the local projections, labor productivity is given by the ratio of GDP to employment, and leverage is defined as the ratio of total assets (TABSNNCB) to total equity, with the latter calculated as total assets minus total liabilities (TLBSNNCB). Table I.1 contains definitions and sources for the data used in the local projection.

[27] https://www.bls.gov/cex/

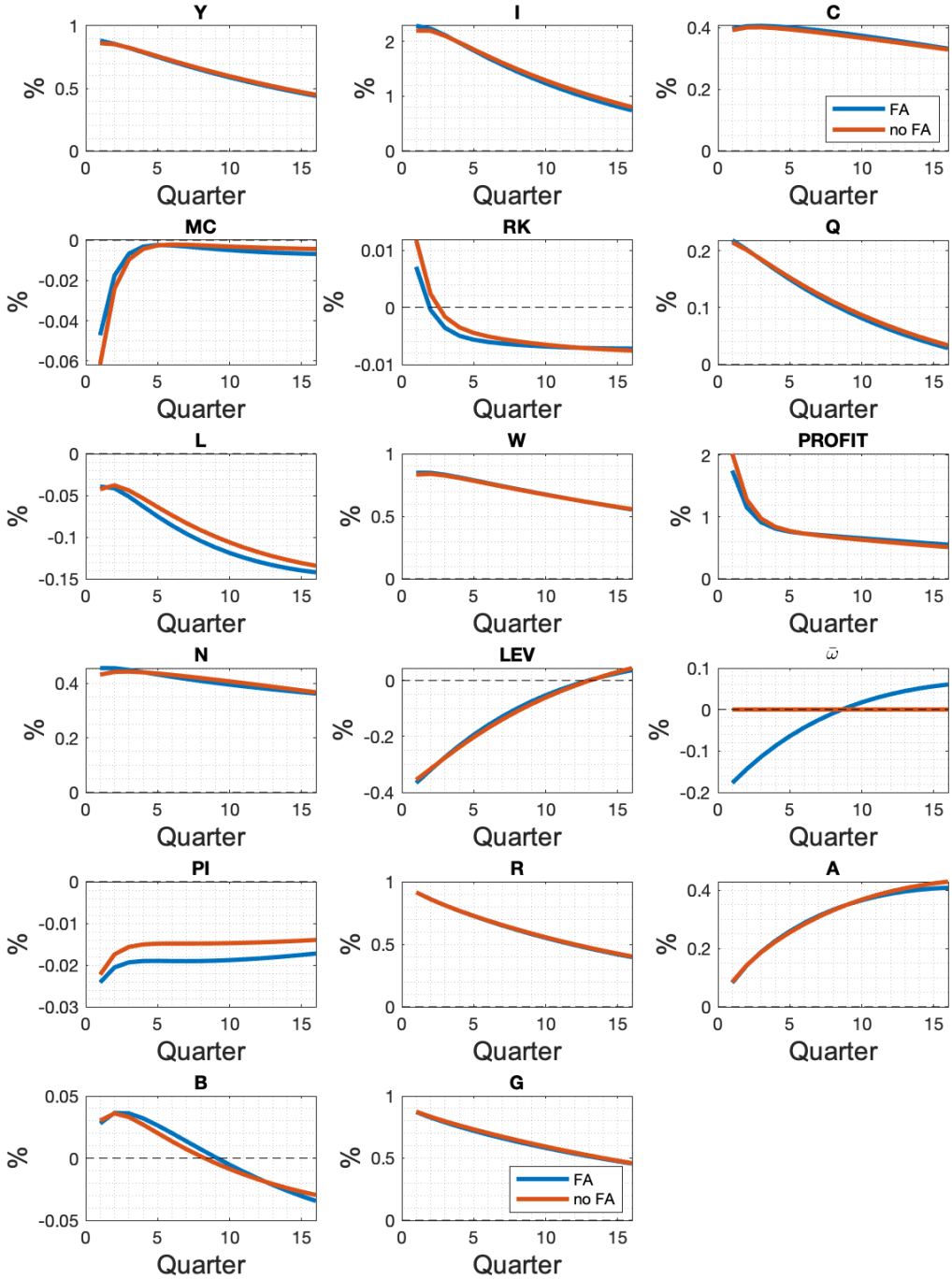

Figure H.1: IRFs for aggregate variables to positive TFP shock

TFP shock $\sigma_z = 0.00915$ with $\rho_z = 0.95$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down.

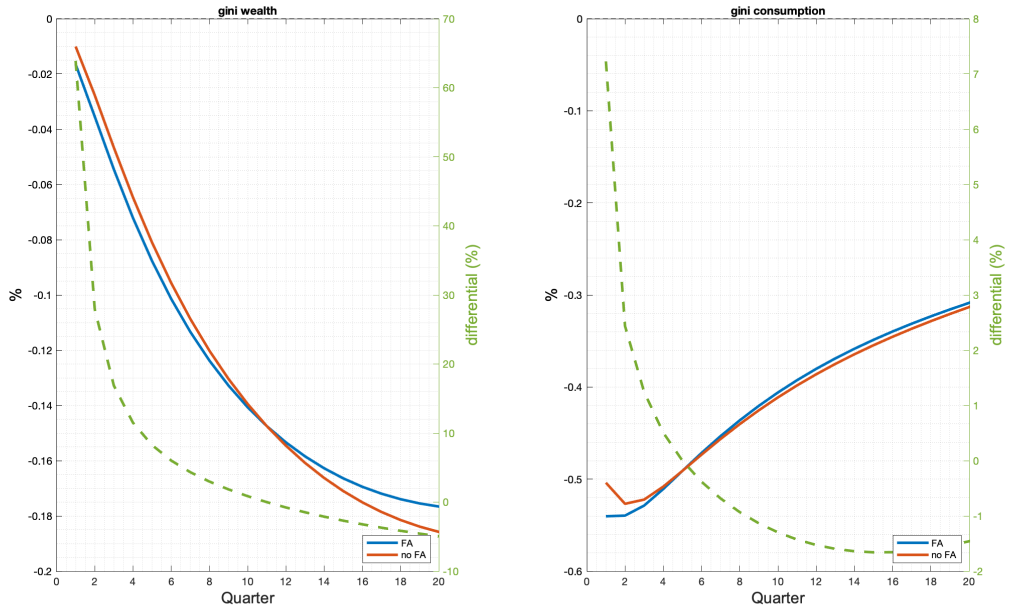

Figure H.2: IRFs for Gini indices to positive TFP shock

The graph on the left-hand side represents fluctuations in the Gini index for wealth, while the one on the right-hand side for consumption. TFP shock $\sigma_z = 0.00915$ with $\rho_z = 0.95$. The blue line refers to an economy with a financial accelerator. The red line refers to the case where financial frictions are shut down. The green dotted line (with values on the right side of the figure) represent the percentage variation from red line to the blue line.

Table I.1: Local projection data

Variable	Definition	Source
CPI	Consumer Price Index for All Urban Consumers	FRED: CPIAUCSL
GDP	Real Gross Domestic Product	FRED: GDPC1
Real Investment	Real Gross Private Domestic Investment	FRED: GPDIC1
Employment	All Employees, Total Nonfarm	FRED: PAYEMS
Charge-off rate	Charge-Off Rate on Business Loans, All Commercial Banks	FRED: CORBLACBS
Consumption	Durable, non-durable, housing services	CEX
s^{GZ}	corporate bond credit spread index	Gilchrist and Zakrajšek (2012)
Leverage	leverage for non-financial corporate firms	FRED: TABSNNCB, TLBSNNCB